JOSÉ RIZAL UNIVERSITY

COLLEGE OF COMPUTER STUDIES & ENGINEERING (CSE)

STUDENT'S CLASS PORTFOLIO

ITC C201 – Database Fundamentals

STUDENT

Abdulrahman Alghannam

YEAR AND SECTION

2nd Year – 201BA

INSTRUCTOR

Virginia B. Loyola, MIT

ACADEMIC YEAR AND SEMESTER

1st Semester, Academic Year 2026–2027

2. Table of Contents

3. Student Profile **3**

4. Course Syllabus / Course Outline **4**

5. Classroom Activities and Exercises **5**

Prelim Activity No. 1 – Conceptual Model (SQL Data Modeler) 6

Prelim Assignment No. 1 – Database Design ERD (hand-drawn) 7

Prelim Activity No. 2 – Supertype and Subtype Entities 8

Midterm Assignment No. 1 – Oracle APEX Account 10

Midterm Activity No. 1 – Creating Tables Using Oracle APEX 11

Midterm Activity No. 2 – Inserting Records 15

Midterm Activity No. 3 – Updating Records 17

Final Activity No. 1 – Alias, Arithmetic and Concatenation Operators 26

Final Seatwork No. 2 – Sorting and Joining Tables 32

6. Laboratory Activities **37**

Midterm Project – Laboratory Activities #2 to #8 38

Laboratory Activities #10 to #12 48

Final Term Activity No. 3 – Joining Tables 64

Final Term Activity No. 4 – SQL Sorting and Joining Tables Part 2 71

7. Projects and Performance Tasks **85**

Prelim Project – ERD and Normalization 86

8. Learning Reflection **88**

9. Course Learning Summary **89**

3. Student Profile

Full Name	Abdulrahman Alghannam
Student Number	26-267514
Program and Year Level	BSIT-BA (Information Technology, Major in Business Analytics) – 2nd Year
Section	201BA
Course Code and Description	ITC C201 – Database Fundamentals

Personal Information

Educational Background

- Bachelor of Science in Information Technology, Major in Business Analytics – José Rizal University (currently in 2nd year, A.Y. 2026–2027)

Interests / Hobbies

- Programming and building small software tools
- Studying quantitative trading and finance
- Mobile gaming (strategy / RPG and MOBA games)

Trainings / Work Experience

Programming skills: Python, C++, pandas, SQL, and Flutter/Dart (self-study and class work).
Hands-on academic and personal projects:

- Flutter + Firebase e-commerce CRUD app (group project, ITC C204)
- Concertify – concert ticket booking website (group project, Human Computer Interaction)
- FoodSave – Food Waste Tracker, a Flutter app aligned with SDG 12 (group project)
- Group digital e-portfolio website for class
- Personal tool for converting scanned Arabic books into searchable text
- Browser-based Monte Carlo simulators built with HTML and JavaScript

Short Personal Introduction

My name is Abdulrahman Alghannam, and I am a second-year BSIT student majoring in Business Analytics at José Rizal University. I am interested in how data can be organized, analyzed, and used to support good decisions. Outside of class, I enjoy programming, and I have experience with Python, C++, pandas, and SQL. I also like to build small tools and projects, such as web-based simulators and group apps made with Flutter. In my free time, I play mobile games and study quantitative trading and finance. My goal is to become a skilled business analyst who can turn data into useful information.

4. Course Syllabus / Course Outline

Summary of the OBE Syllabus given by the instructor (IT Department, 1st Semester SY 2026–2027)

Course	ITC C201 – Database Fundamentals (3 units: 2 hours lecture, 3 hours laboratory)	**Placement**	2nd Year, 1st Semester
Instructor	Virginia B. Loyola, MIT	**Pre-requisite**	MAT C604

Course Description

This course introduces database concepts, design, creation, use, and management. It covers data dictionaries, normalization, data integrity, data modeling, and the creation of simple tables, queries, reports, and forms. Students use Oracle Database SQL statements for data definition, data manipulation, and data queries, and the course ends with a project in which students design and demonstrate a normalized database solution for a business organization.

Course Learning Outcomes

- **CLO1:** Define the concepts and applications of database, database models, database management systems, and relational database theory.
- **CLO2:** Identify and apply data modeling concepts to a business problem, convert the logical model into a physical model, and apply normalization.
- **CLO3:** Use basic SQL statements to create, manipulate, and query a database, and build one working system using a relational database and a front-end language.

Course Outline

Period	Main topics	Major outputs
Prelim (Weeks 1–6)	Class orientation and OBE; database fundamentals (data vs. information, types of data and DBMS, data models); entity relationship modeling and ERDs; enhanced ER modeling (supertypes and subtypes); Oracle SQL Data Modeler	Seatwork, assignments, lab activity on logical and relational design, Prelim Project, Prelim Exam
Midterm (Weeks 7–12)	Creating an Oracle APEX account, databases, and tables; normalization (1NF, 2NF, 3NF); ALTER and DROP statements; SQL constraints (primary key, not null, unique, check, foreign key); INSERT, UPDATE, and DELETE statements	Laboratory activities on DDL and DML, Midterm Project, Midterm Exam
Final (Weeks 13–18)	SQL operators (WHERE, AND, OR, NOT), alias, LIKE, wildcards, IN, BETWEEN; group functions; inner, outer, self, cross, and natural joins	Laboratory activities on operators and joining tables, Final Group Project (e-commerce application with database), e-portfolio, Final Exam

Grading System

Term	Weight	Components
Prelim	30%	Class standing 70% (written works and quizzes, performance tasks and lab activities, project) + Major exam 30%
Midterm	30%	Same structure as Prelim
Final	40%	Same structure as Prelim

Course Requirements and References

Requirements: quizzes, major examinations, case analysis and laboratory exercises, discussion boards, and portfolio/research/project.

Textbook: Kuzmiakova, A. (Ed.). (2023). *The Creation and Management of Database Systems*. Arcler Press. **Other references:** Salter, K. (2024), *Fundamentals of Database Systems*; Weiss, K. A., and Balti, H. (2023), *Job Ready SQL*; Oracle Database Documentation; W3Schools SQL Tutorial.

SECTION 5

Classroom Activities and Exercises

Contents of this section

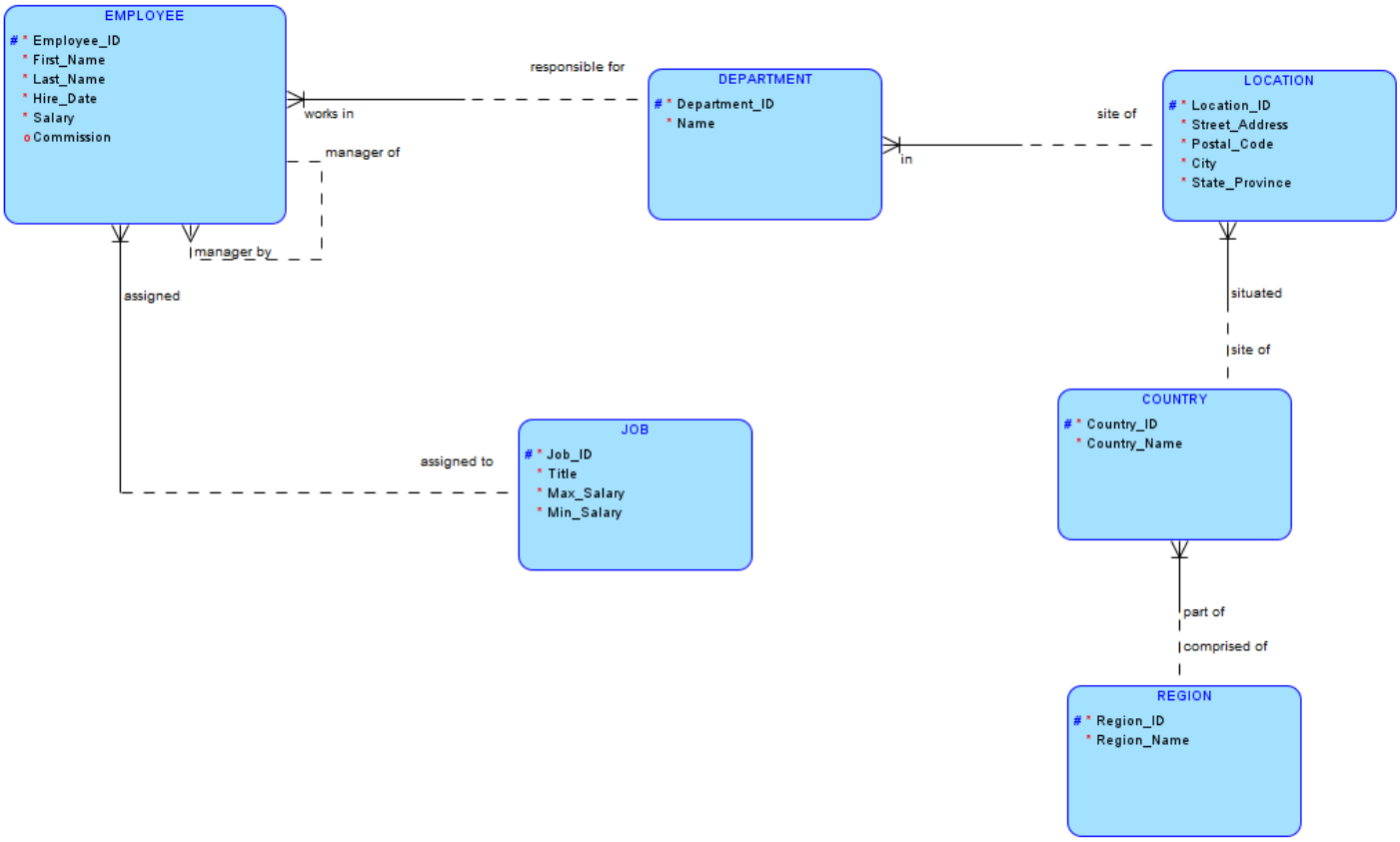
- Prelim Activity No. 1 – Conceptual Model (SQL Data Modeler)
- Prelim Assignment No. 1 – Database Design ERD (hand-drawn)
- Prelim Activity No. 2 – Supertype and Subtype Entities
- Midterm Assignment No. 1 – Oracle APEX Account
- Midterm Activity No. 1 – Creating Tables Using Oracle APEX
- Midterm Activity No. 2 – Inserting Records
- Midterm Activity No. 3 – Updating Records
- Final Activity No. 1 – Alias, Arithmetic and Concatenation Operators
- Final Seatwork No. 2 – Sorting and Joining Tables

TITLE: PRELIM ACTIVITY NO.1 – CONCEPTUAL MODEL USING SQL DATA MODELER

PREPARED BY: <Abdulrahman alghannam>

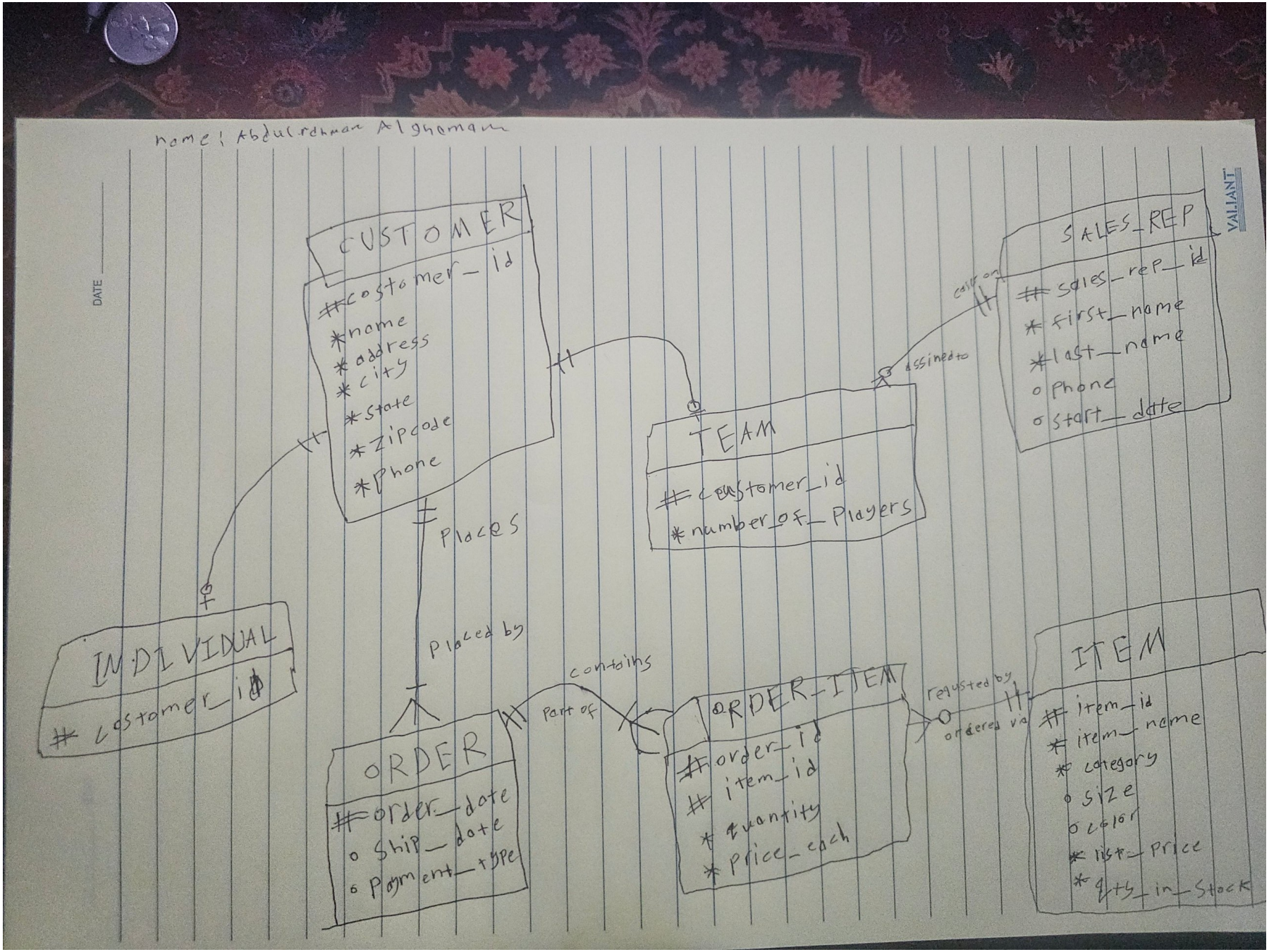
ITC C201 – 201BA

SCREENSHOT OF YOUR CONCEPTUAL MODEL (ERD) – FILE DD 2.3 SLIDE 13



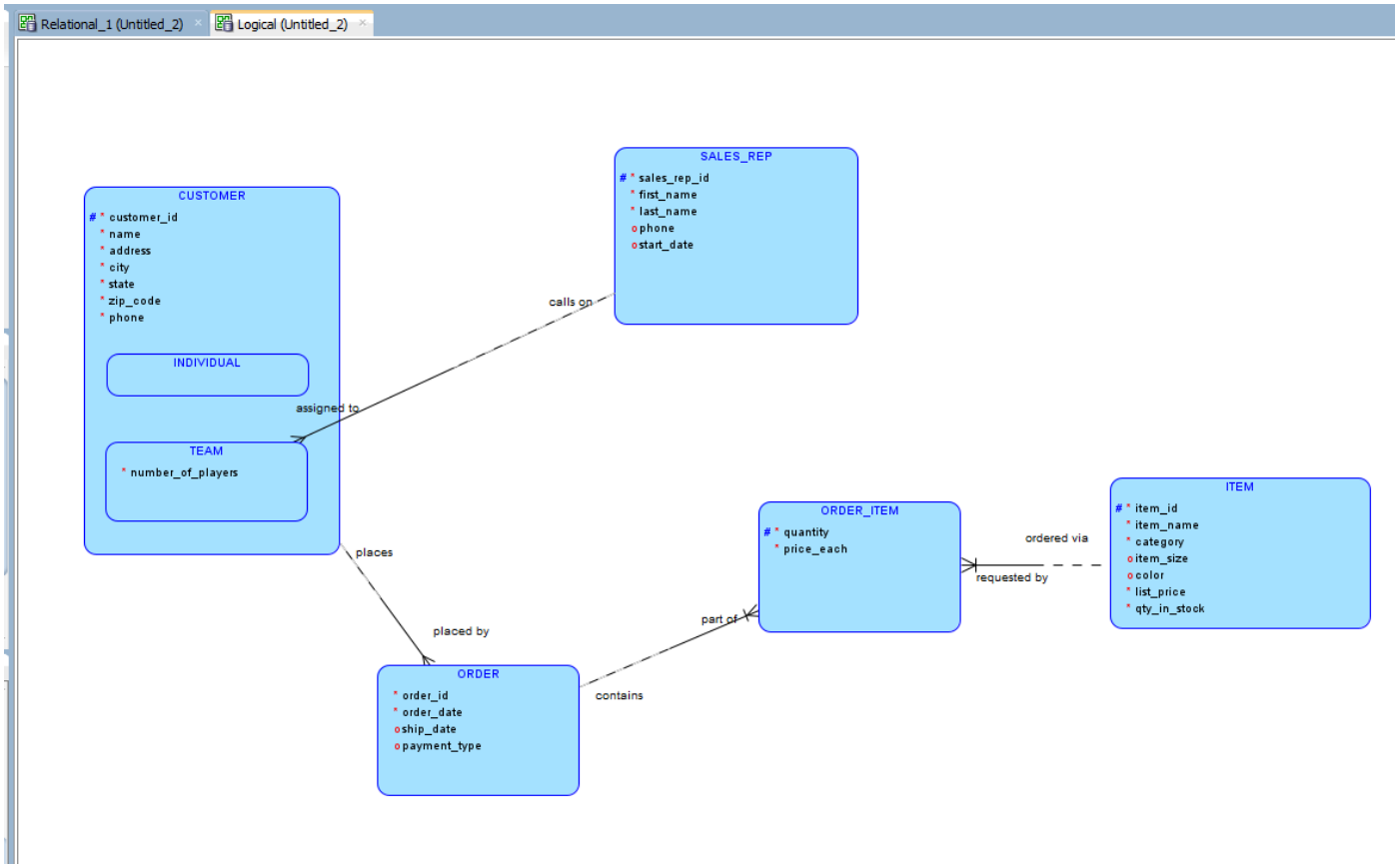
Prelim Assignment No. 1 – Database Design ERD

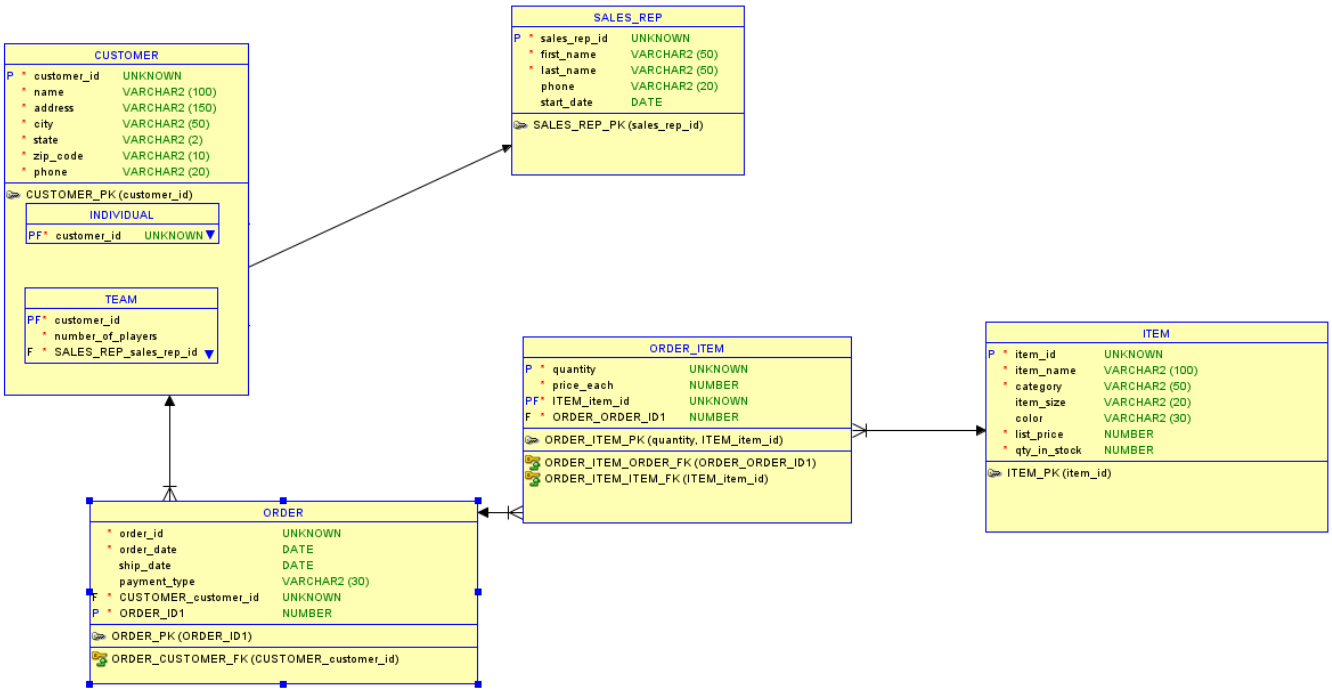
ITC C201 – 201BA | Abdulrahman Alghannam | Hand-drawn ERD



It c201- 201BA

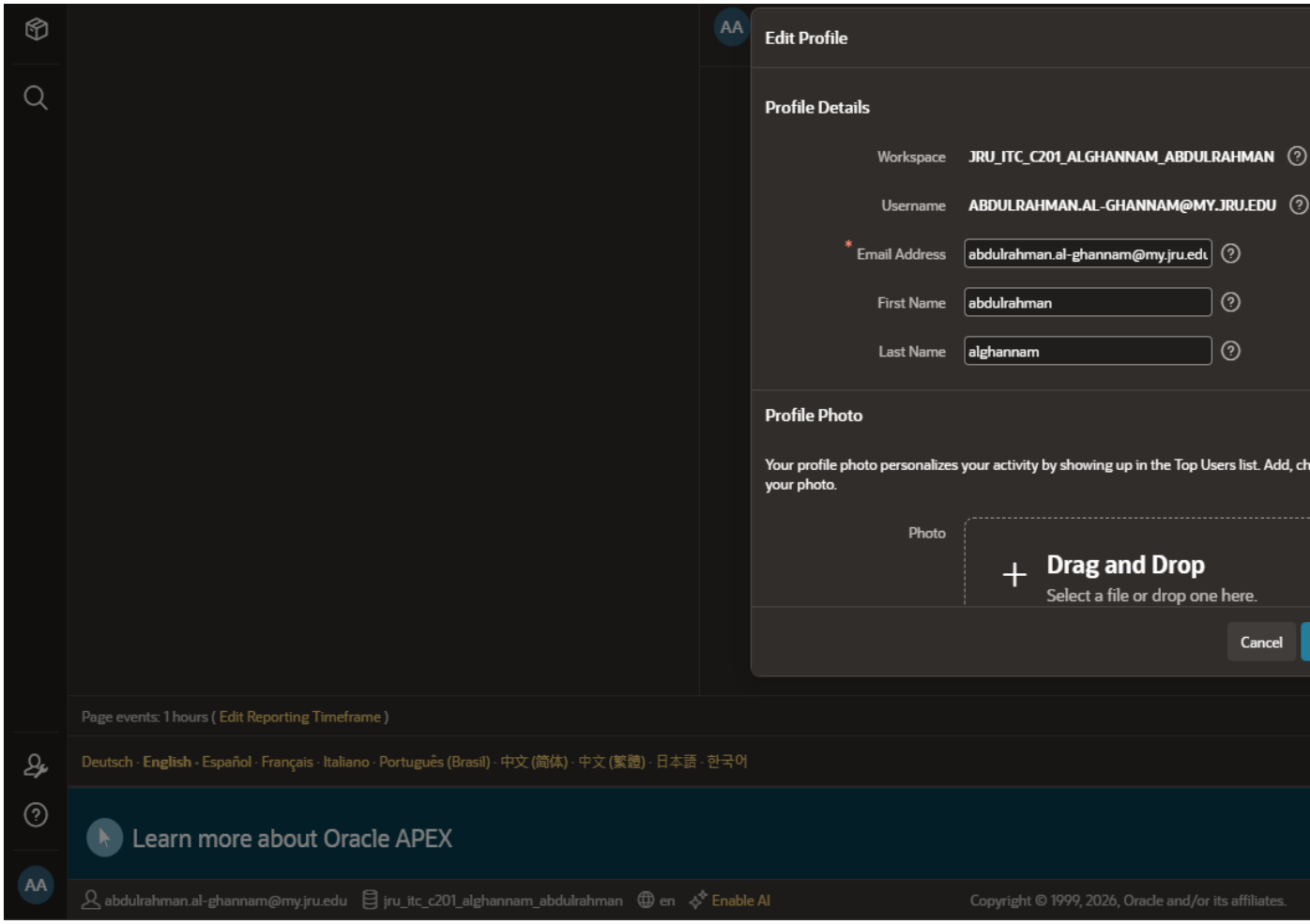
Name : Abdulrahman alghannam




CUSTOMER
P * customer_id UNKNOWN
* name VARCHAR2 (100)
* address VARCHAR2 (150)
* city VARCHAR2 (50)
* state VARCHAR2 (2)
* zip_code VARCHAR2 (10)
* phone VARCHAR2 (20)
CUSTOMER_PK (customer_id)
INDIVIDUAL
PF * customer_id UNKNOWN
TEAM
PF * customer_id
* number_of_players
F * SALES_REP_sales_rep_id
SALES_REP
P * sales_rep_id UNKNOWN
* first_name VARCHAR2 (50)
* last_name VARCHAR2 (50)
phone VARCHAR2 (20)
start_date DATE
SALES_REP_PK (sales_rep_id)
ORDER_ITEM
P * quantity UNKNOWN
* price_each NUMBER
PF * ITEM_item_id UNKNOWN
F * ORDER_ORDER_ID1 NUMBER
ORDER_ITEM_PK (quantity, ITEM_item_id)
ORDER_ITEM_ORDER_FK (ORDER_ORDER_ID1)
ORDER_ITEM_ITEM_FK (ITEM_item_id)
ITEM
P * item_id UNKNOWN
* item_name VARCHAR2 (100)
* category VARCHAR2 (50)
item_size VARCHAR2 (20)
color VARCHAR2 (30)
* list_price NUMBER
* qty_in_stock NUMBER
ITEM_PK (item_id)
ORDER
* order_id UNKNOWN
* order_date DATE
ship_date DATE
payment_type VARCHAR2 (30)
F * CUSTOMER_customer_id UNKNOWN
P * ORDER_ID1 NUMBER
ORDER_PK (ORDER_ID1)
ORDER_CUSTOMER_FK (CUSTOMER_customer_id)

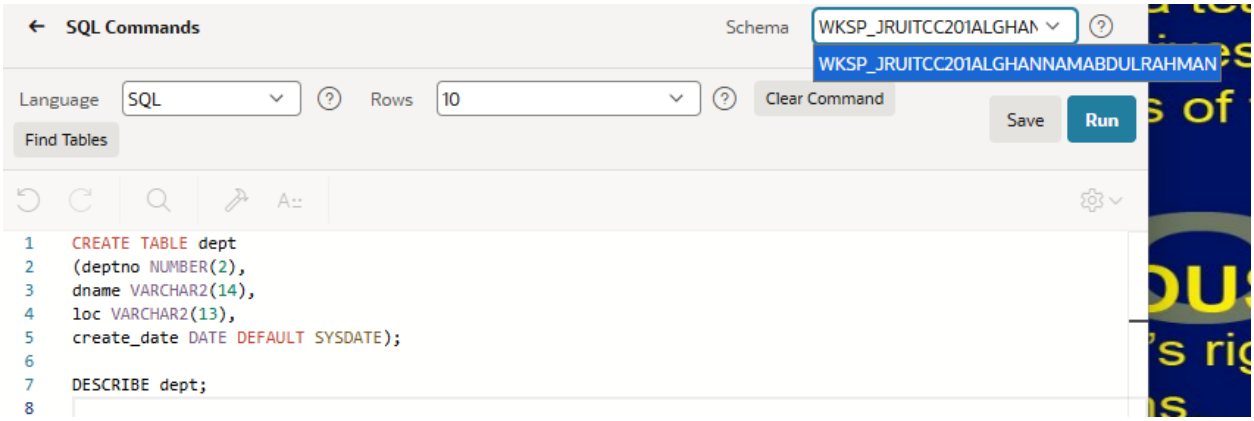
Midterm Assignment No. 1 – Oracle APEX Account

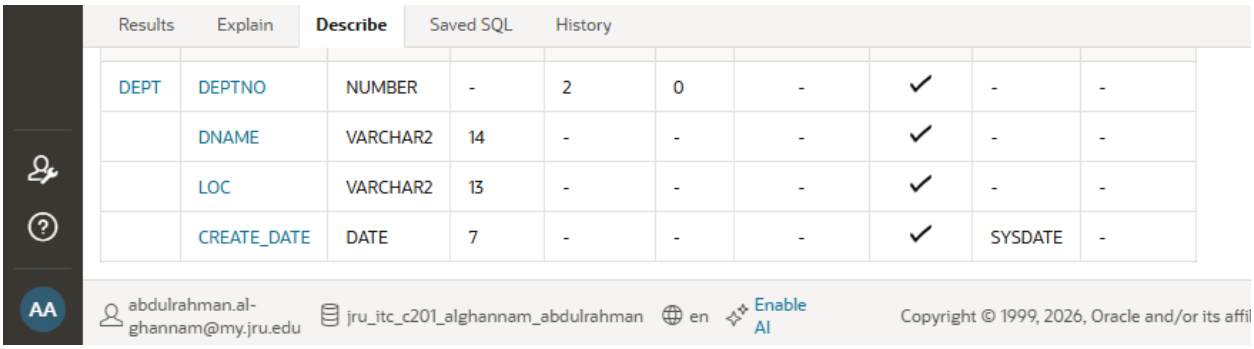
Name: Alghannam Abdulrahman

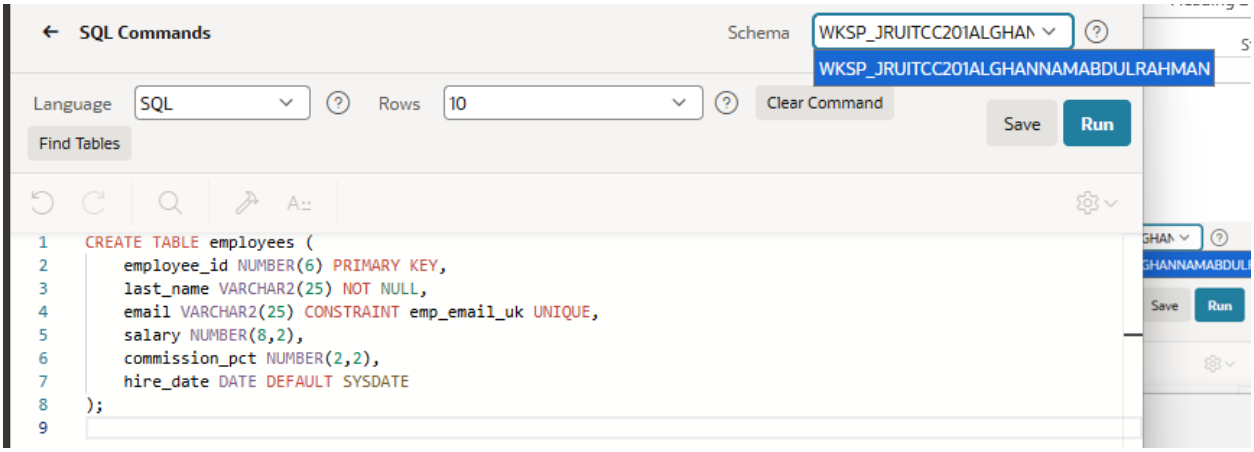


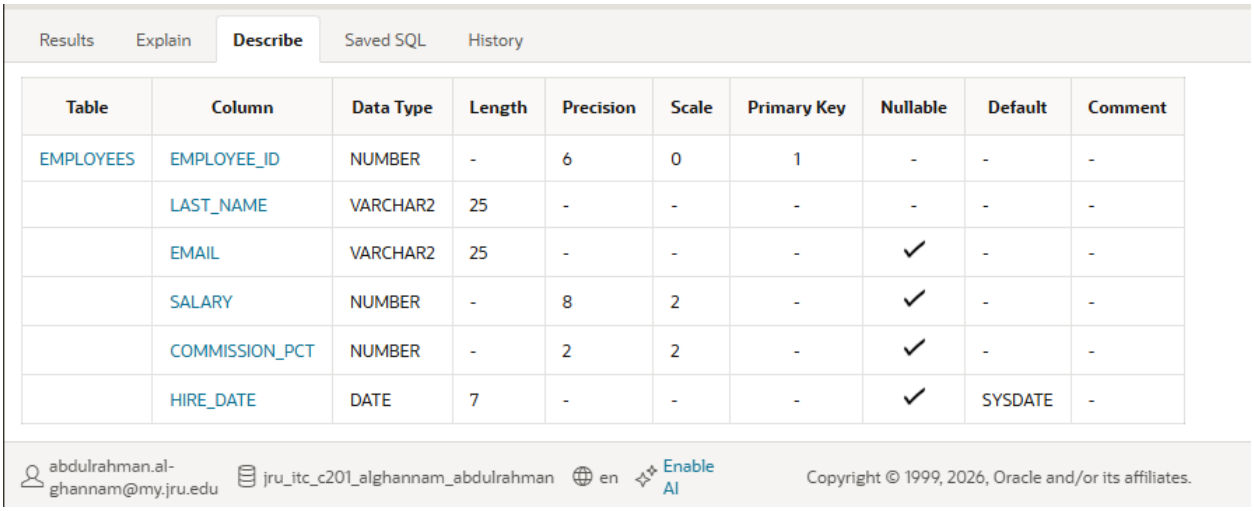
Name: abdulrahman alghannam

MIDTERM ACTIVITY NO.1 _ CREATING TABLE USING ORACLE APEX



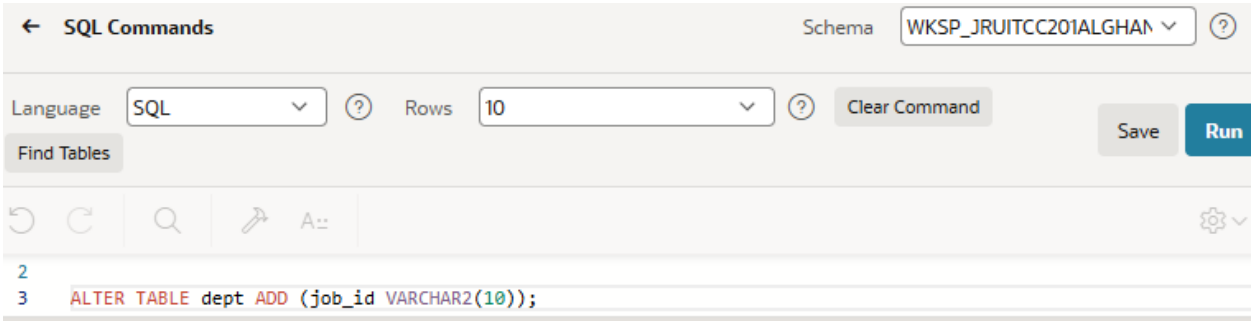


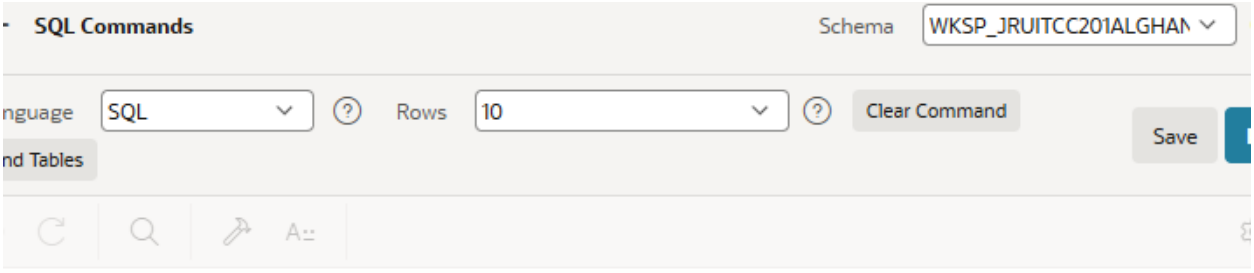




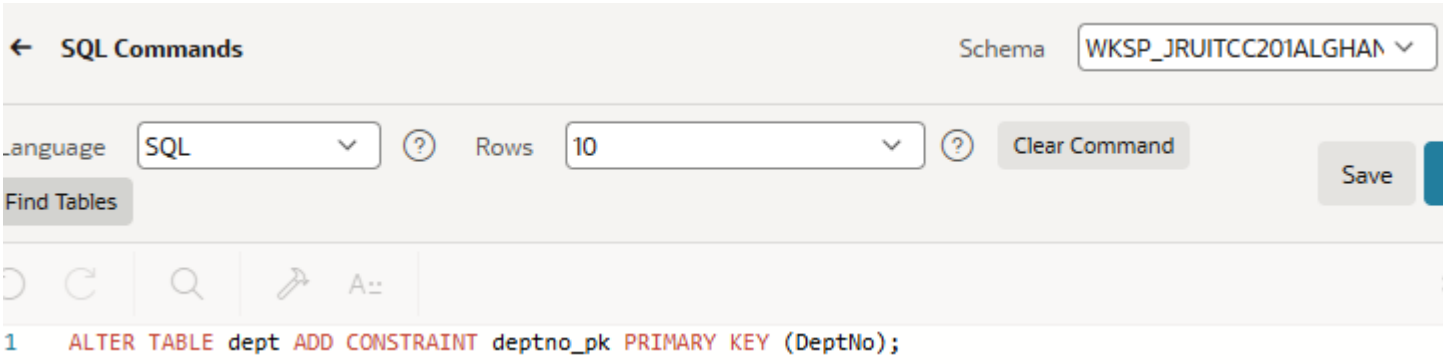
Results | Explain | **Describe** | Saved SQL | History

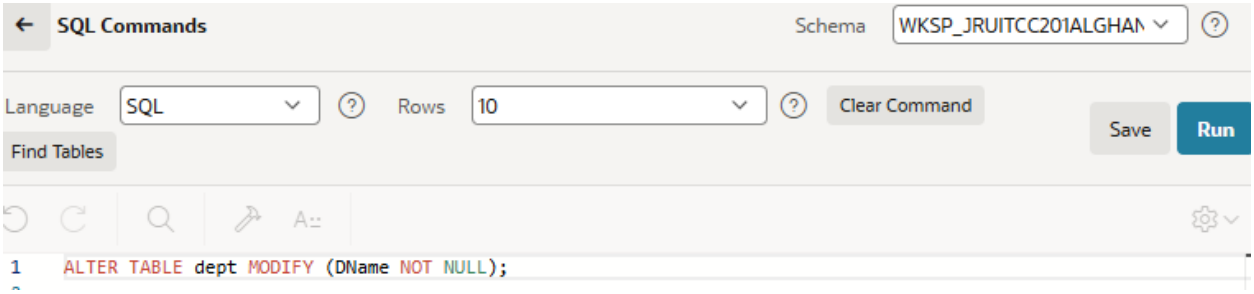
Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMPLOYEES	EMPLOYEE_ID	NUMBER	-	6	0	1	-	-	-
	LAST_NAME	VARCHAR2	25	-	-	-	-	-	-
	EMAIL	VARCHAR2	25	-	-	-	✓	-	-
	SALARY	NUMBER	-	8	2	-	✓	-	-
	COMMISSION_PCT	NUMBER	-	2	2	-	✓	-	-
	HIRE_DATE	DATE	7	-	-	-	✓	SYSDATE	-




SQL Commands
Schema
WKSP_JRUITCC201ALGHAN
SQL
Rows
10
Clear Command
Save
ALTER TABLE employees ADD (job_id VARCHAR2(10));


SQL Commands
Schema
WKSP_JRUITCC201ALGHAN
Language
SQL
Rows
10
Clear Command
Save
Find Tables
1 ALTER TABLE dept MODIFY (Dname VARCHAR2(30));


SQL Commands
Schema
WKSP_JRUITCC201ALGHAN
Language
SQL
Rows
10
Clear Command
Save
Find Tables
1 ALTER TABLE dept ADD CONSTRAINT deptno_pk PRIMARY KEY (DeptNo);


SQL Commands
Schema
WKSP_JRUITCC201ALGHAN
Language
SQL
Rows
10
Clear Command
Save
Run
Find Tables
1 ALTER TABLE dept MODIFY (DName NOT NULL);

Results | Explain | **Describe** | Saved SQL | History

Object Type **TABLE** ⓘ Object **DEPT** ⓘ

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
DEPT	DEPTNO	NUMBER	-	2	0	1	-	-	-
	DNAME	VARCHAR2	30	-	-	-	-	-	-
	LOC	VARCHAR2	13	-	-	-	✓	-	-
	CREATE_DATE	DATE	7	-	-	-	✓	SYSDATE	-
	JOB_ID	VARCHAR2	10	-	-	-	✓	-	-

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Results | Explain | **Describe** | Saved SQL | History

EMPLOYEES	EMPLOYEE_ID	NUMBER	-	6	0	1	-	-	-
	LAST_NAME	VARCHAR2	25	-	-	-	-	-	-
	EMAIL	VARCHAR2	25	-	-	-	✓	-	-
	SALARY	NUMBER	-	8	2	-	✓	-	-
	COMMISSION_PCT	NUMBER	-	2	2	-	✓	-	-
	HIRE_DATE	DATE	7	-	-	-	✓	SYSDATE	-
	JOB_ID	VARCHAR2	10	-	-	-	✓	-	-

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```sql
INSERT INTO copy_dept (deptno, dname, loc)
VALUES (10, 'administration', 1700)
,(20, 'Marketing', 1800)
, (30, 'Purchasing', 1700)
, (40, 'Human Resources', 2400)
, (50, 'Shipping', 1500)
, (60, 'IT', 1400)
, (70, 'Public Relations', 2700)
, (80, 'Sales', 1500);

SELECT * FROM copy_dept;
```

Results | Explain | Describe | Saved SQL | History

DEPTNO	DNAME
10	administration
10	administration
20	Marketing
30	Purchasing
40	Human Resources
50	Shipping
60	IT
70	Public Relations
80	Sales

9 rows returned in 0.01 seconds Download

```sql
INSERT INTO copy_employees (employee_id, last_name, email, salary)
VALUES
    (100, 'King', 'SKING', 24000),
    (101,'Kochhar', 'NKOCHHAR', 17000),
    (102, 'De Haan', 'LDEHAAN', 17000),
    (200, 'Whalen', 'JWHALEN', 4400);

    SELECT * FROM copy_employees;
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY
100	King	SKING	24000
101	Kochhar	NKOCHHAR	17000
102	De Haan	LDEHAAN	17000
200	Whalen	JWHALEN	4400

4 rows returned in 0.01 seconds Download

Name : abdulrahman al-ghannam

201BA

```
1  ALTER TABLE COPY_EMPLOYEES
2  ADD (FirstName    VARCHAR2(15),
3       DepartmentID NUMBER(2),
4       Manager_ID   NUMBER(3));
5
```

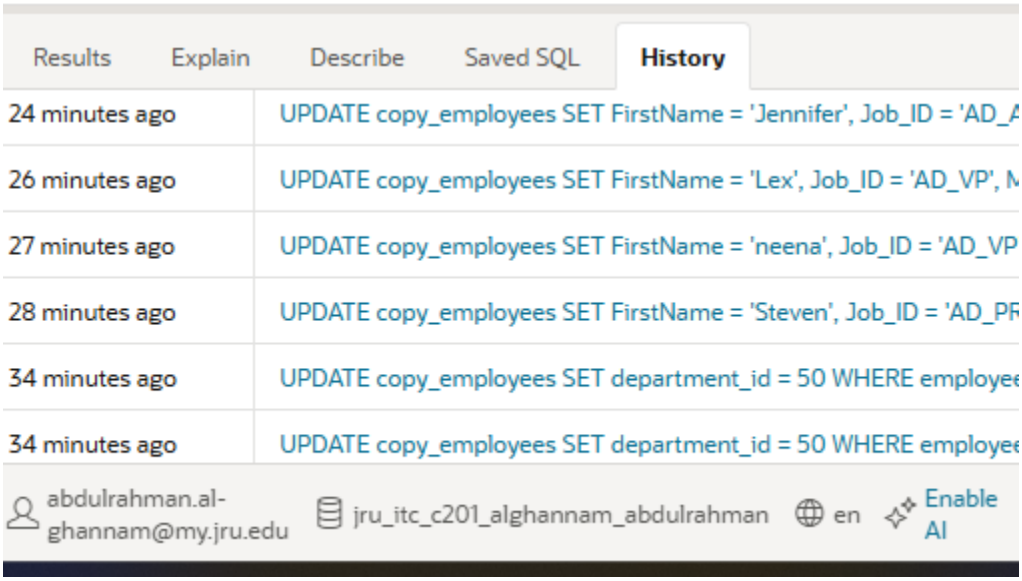
Results | Explain | Describe | Saved SQL | History

```
Table altered.
```

0.13 seconds

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```sql
UPDATE copy_employees
SET    FirstName    = 'Steven',
       Job_ID       = 'AD_PRES',
       MANAGER_ID   = NULL,
       DepartmentID = 60
WHERE  employee_id  = 100;
```



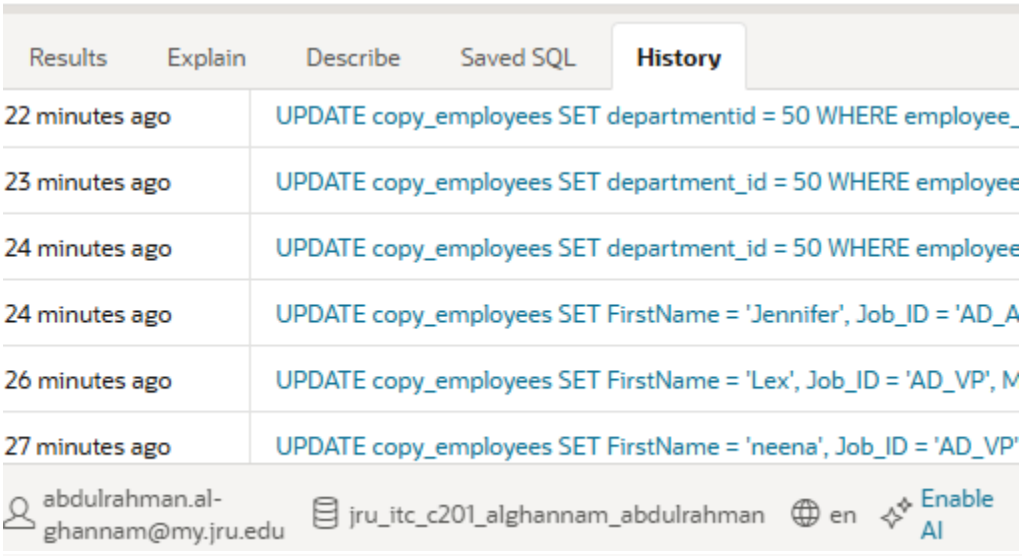
```sql
UPDATE copy_employees
SET    FirstName    = 'neena',
       Job_ID       = 'AD_VP',
       MANAGER_ID   = 100,
       DepartmentID = 60
WHERE  employee_id  = 101;
```

Results | Explain | Describe | Saved SQL | **History**

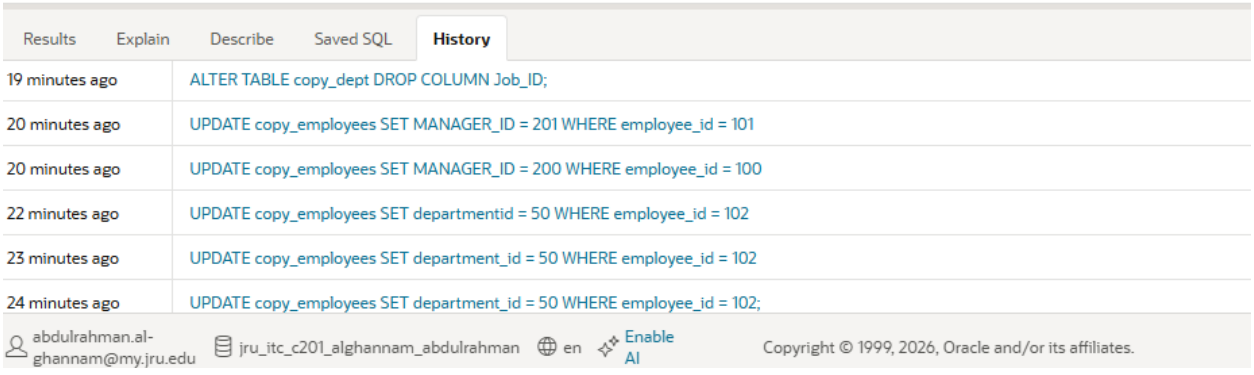
24 minutes ago	UPDATE copy_employees SET FirstName = 'Jennifer', Job_ID = 'AD_ASS
26 minutes ago	UPDATE copy_employees SET FirstName = 'Lex', Job_ID = 'AD_VP', MAN
27 minutes ago	UPDATE copy_employees SET FirstName = 'neena', Job_ID = 'AD_VP', M
28 minutes ago	UPDATE copy_employees SET FirstName = 'Steven', Job_ID = 'AD_PRES',
34 minutes ago	UPDATE copy_employees SET department_id = 50 WHERE employee_id
34 minutes ago	UPDATE copy_employees SET department_id = 50 WHERE employee_id

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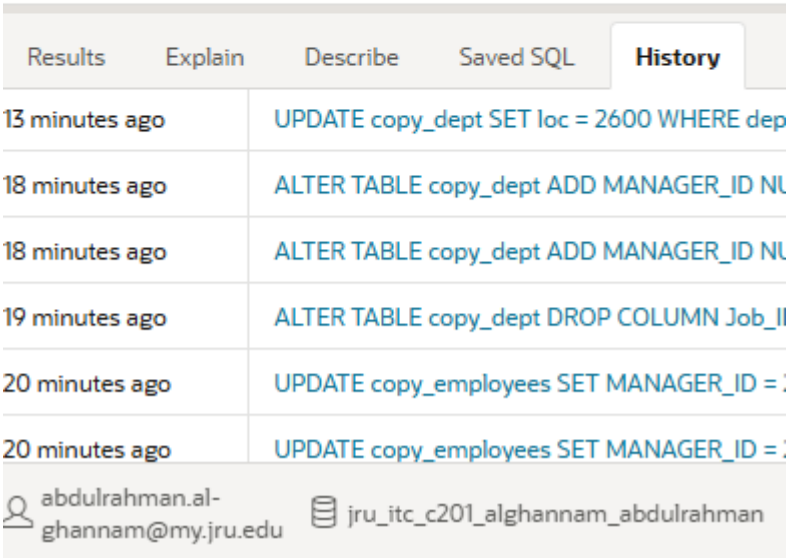
```sql
UPDATE copy_employees
SET    FirstName    = 'Lex',
       Job_ID       = 'AD_VP',
       MANAGER_ID   = 100,
       DepartmentID = 60
WHERE  employee_id  = 102;
```



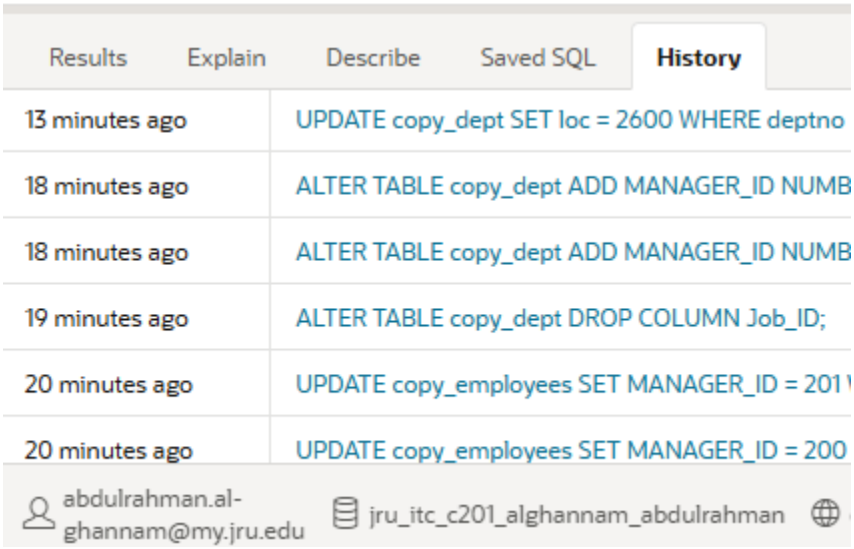

```sql
UPDATE copy_employees SET departmentid = 50 WHERE employee_id = 102
```



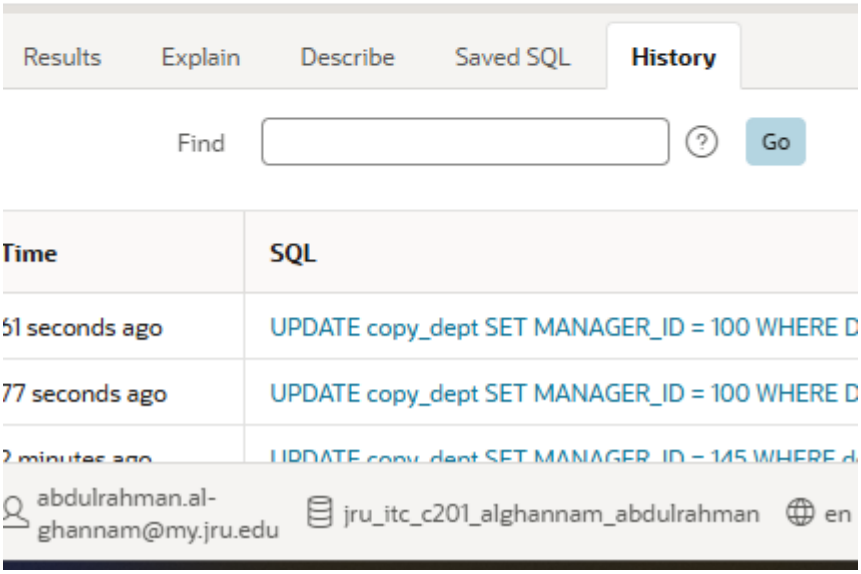
```
1   ALTER TABLE copy_dept
2   DROP COLUMN Job_ID;
```



```
1   ALTER TABLE copy_dept
2   ADD MANAGER_ID NUMBER(3);
```



```
1   UPDATE copy_dept
2   SET    MANAGER_ID = 100
3   WHERE  DNAME = 'Purchasing';
```

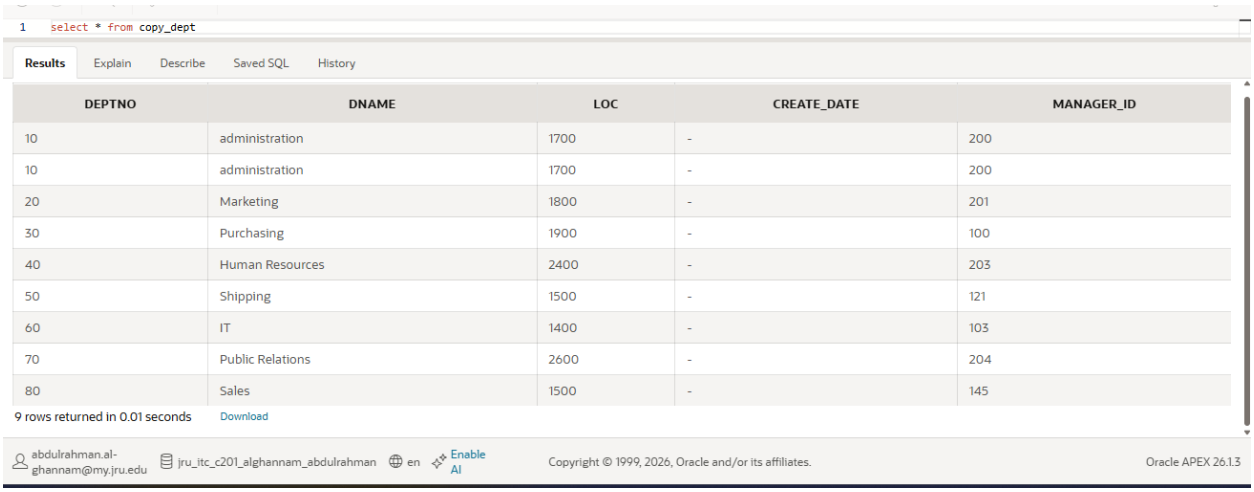



```
1   select * from copy_employees
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	JOB_ID	FIRSTNAME	DEPARTMENTID	MANAGER_ID
100	King	SKING	24000	-	-	AD_PRES	Steven	60	200
101	Kochhar	NKOCHHAR	17000	-	-	AD_VP	neena	60	201
102	De Haan	LDEHAAN	17000	-	-	AD_VP	Lex	50	100
200	Whalen	JWHALEN	4400	-	-	AD_ASST	Jennifer	10	101

4 rows returned in 0.02 seconds Download


1 select * from copy_dept
Results
Explain
Describe
Saved SQL
History
DEPTNO
DNAME
LOC
CREATE_DATE
MANAGER_ID
10 administration 1700 - 200
10 administration 1700 - 200
20 Marketing 1800 - 201
30 Purchasing 1900 - 100
40 Human Resources 2400 - 203
50 Shipping 1500 - 121
60 IT 1400 - 103
70 Public Relations 2600 - 204
80 Sales 1500 - 145
9 rows returned in 0.01 seconds
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Oracle APEX 26.1.3

```sql
SELECT LAST_NAME, SALARY+300 as "UpdateSalary", salary*12 as "AnnualSalry" FROM COPY_EMPLOYEES;
```

Results | Explain | Describe | Saved SQL | History

LAST_NAME	UpdateSalary	AnnualSalry
King	24300	288000
Kochhar	17300	204000
De Haan	17300	204000

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```sql
ALTER TABLE COPY_EMPLOYEES
MODIFY (COMMISSION_PCT NUMBER(10,2));


update COPY_EMPLOYEES set COMMISSION_PCT = SALARY * 0.10;

SELECT * FROM COPY_EMPLOYEES;
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	JOB_ID	FIRSTNAME	DEPARTMENTID	MANAGER_ID
100	King	SKING	24000	2400	-	AD_PRES	Steven	60	200
101	Kochhar	NKOCHHAR	17000	1700	-	AD_VP	neena	60	201
102	De Haan	LDEHAAN	17000	1700	-	AD_VP	Lex	50	100
200	Whalen	JWHALEN	4400	440	-	AD_ASST	Jennifer	10	101

4 rows returned in 0.00 seconds Download

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```
10
11
12  SELECT last_name || ', ' ||firstname as FullName FROM COPY_EMPLOYEES;
13
14
15
16
```

Results | Explain | Describe | Saved SQL | History

FULLNAME
King, Steven
Kochhar, neena
De Haan, Lex
Whalen, Jennifer

4 rows returned in 0.01 seconds Download

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```
15
16  SELECT last_name || q'['s employees id : ]' ||EMPLOYEE_id as EMPLOYEES_LastName_ID   FROM COPY_EMPLOYEES;
17
18
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEES_LASTNAME_ID
King's employees id : 100
Kochhar's employees id : 101
De Haan's employees id : 102
Whalen's employees id : 200

4 rows returned in 0.00 seconds Download

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```sql
UPDATE copy_employees
SET hire_date = TO_DATE('17-06-03', 'DD-MM-RR')
WHERE employee_id = 100

UPDATE copy_employees
SET hire_date = TO_DATE('21-09-05', 'DD-MM-RR')
WHERE employee_id = 101

UPDATE copy_employees
SET hire_date = TO_DATE('13-01-01', 'DD-MM-RR')
WHERE employee_id = 102

select * from copy_employees
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE
106	Pataballa	-	4800	-	9/21/2005
100	King	SKING	24000	2400	6/17/2003
101	Kochhar	NKOCHHAR	17000	1700	9/21/2005
102	De Haan	LDEHAAN	17000	1700	1/13/2001
200	Whalen	JWHALEN	4400	440	9/21/2005

```
WHERE employee_id = 101

UPDATE copy_employees
 SET departmentid = 40
 WHERE employee_id = 102

UPDATE copy_employees
 SET departmentid = 30
 WHERE employee_id = 103

UPDATE copy_employees
 SET departmentid = 20
 WHERE employee_id = 104

UPDATE copy_employees
 SET departmentid = 30
 WHERE employee_id = 105


SELECT * FROM copy_employees
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	JOB_ID	FIRSTNAME	DEPARTMENTID
100	King	SKING	24000	2400	6/17/2003	AD_PRES	Steven	10
101	Kochhar	NKOCHHAR	17000	1700	9/21/2005	AD_VP	neena	20
102	De Haan	LDEHAAN	17000	1700	1/13/2001	AD_VP	Lex	40
103	Hunold	-	9000	-	9/21/2005	-	-	30
104	Ernst	-	6000	-	9/21/2005	-	-	20
105	Austin	-	4800	-	9/21/2005	-	-	30

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```
27
28  ALTER TABLE copy_employees
29
30  ADD (UpdatedSalary
31
32  NUMBER(10, 2))
33
34  desc copy_employees
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
```

Results | Explain | **Describe** | Saved SQL | History

	HIRE_DATE	DATE	7	-	-	-	✓
	JOB_ID	VARCHAR2	10	-	-	-	✓
	FIRSTNAME	VARCHAR2	15	-	-	-	✓
	DEPARTMENTID	NUMBER	-	2	0	-	✓
	MANAGER_ID	NUMBER	-	3	0	-	✓
	DEPTID	NUMBER	22	-	-	-	✓
	UPDATEDSALARY	NUMBER	-	10	2	-	✓

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```
34
35  UPDATE copy_employees
36  SET UpdatedSalary = salary + 300
37
38
39
40  SELECT employee_id , salary , UPDATEDSALARY FROM copy_employees;
41
42
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	SALARY	UPDATEDSALARY
103	9000	9300
104	6000	6300
105	4800	5100
106	4800	5100
100	24000	24300
101	17000	17300
102	17000	17300
200	4400	4700

8 rows returned in 0.01 seconds Download

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5.1 sorting table

```
1  desc copy_employees
2
3  select last_name , DEPARTMENTID , HIRE_DATE
4  from copy_employees
5  order by HIRE_DATE;
```

Results | Explain | Describe | Saved SQL | History

LAST_NAME	DEPARTMENTID	HIRE_DATE
De Haan	40	1/13/2001
King	10	6/17/2003
Austin	30	9/21/2005
Whalen	30	9/21/2005
Kochhar	20	9/21/2005
Ernst	20	9/21/2005
Hunold	30	9/21/2005

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```
2
3  select last_name , DEPARTMENTID , HIRE_DATE
4  from copy_employees
5  order by HIRE_DATE desc;
```

Results | Explain | Describe | Saved SQL | History

LAST_NAME	DEPARTMENTID	HIRE_DATE
Hunold	30	9/21/2005
Ernst	20	9/21/2005
Austin	30	9/21/2005
Whalen	30	9/21/2005
Pataballa	-	9/21/2005
Kochhar	20	9/21/2005
King	10	6/17/2003

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```
1  desc copy_employees
2
3  select last_name , salary , salary * 0.2 bonus
4  from copy_employees
5  order by bonus desc;
```

Results | Explain | Describe | Saved SQL | History

LAST_NAME	SALARY	BONUS
King	24000	4800
Kochhar	17000	3400
De Haan	17000	3400
Hunold	9000	1800
Ernst	6000	1200
Austin	4800	960
Pataballa	4800	960

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```
3  select rownum as "longeest employed" , last_name  , hire_date
4  from (select last_name , HIRE_DATE
5  from copy_employees
6  order by HIRE_DATE)
7  where rownum <= 3
```

Results | Explain | Describe | Saved SQL | History

longeest employed	LAST_NAME	HIRE_DATE
1	De Haan	1/13/2001
2	King	6/17/2003
3	Hunold	9/21/2005

rows returned in 0.00 seconds Download

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```sql
select last_name , salary , DEPARTMENTID
from copy_employees
order by DEPARTMENTID , salary desc;
```

Results | Explain | Describe | Saved SQL | History

LAST_NAME	SALARY	DEPARTMENTID
King	24000	10
Kochhar	17000	20
Ernst	6000	20
Hunold	9000	30
Austin	4800	30
Whalen	4400	30

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Joining table

```
4
5   SELECT e.employee_id, e.last_name, d.deptno, d.dname
6   FROM copy_employees e
7   JOIN copy_dept d ON (e.deptno = d.deptno)
8   and e.deptno = 10;
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	DEPTNO	DNAME
100	King	10	administration

1 rows returned in 0.01 seconds Download

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Left

```
4
5   SELECT e.employee_id, e.last_name, d.deptno, d.dname
6   FROM copy_employees e left outer JOIN copy_dept d
7   ON (e.deptno = d.deptno)
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	DEPTNO	DNAME
100	King	10	administration
104	Ernst	20	Marketing
101	Kochhar	20	Marketing
103	Hunold	30	Purchasing
105	Austin	30	Purchasing
200	Whalen	30	Purchasing
102	De Haan	40	Human Resources
106	Pataballa	-	-

8 rows returned in 0.01 seconds Download

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Right

```
4
5   SELECT e.employee_id, e.last_name, d.deptno, d.dname
6   FROM copy_employees e right outer JOIN copy_dept d
7   ON (e.deptno = d.deptno)
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	DEPTNO	DNAME
100	King	10	administration
104	Ernst	20	Marketing
101	Kochhar	20	Marketing
105	Austin	30	Purchasing
200	Whalen	30	Purchasing
103	Hunold	30	Purchasing
102	De Haan	40	Human Resources
-	-	50	Shipping
-	-	60	IT

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Full

```
5   SELECT e.employee_id, e.last_name, d.deptno, d.dname
6   FROM copy_employees e full outer JOIN copy_dept d
7   ON (e.deptno = d.deptno)
8
```

Results | Explain | Describe | Saved SQL | History

EMPLOYEE_ID	LAST_NAME	DEPTNO	DNAME
104	Ernst	20	Marketing
101	Kochhar	20	Marketing
103	Hunold	30	Purchasing
105	Austin	30	Purchasing
200	Whalen	30	Purchasing
102	De Haan	40	Human Resources
-	-	50	Shipping
-	-	60	IT
-	-	70	Public Relations

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SECTION 6

Laboratory Activities

Contents of this section

- Midterm Project – Laboratory Activities #2 to #8
- Laboratory Activities #10 to #12
- Final Term Activity No. 3 – Joining Tables
- Final Term Activity No. 4 – SQL Sorting and Joining Tables Part 2

Laboratory Activity #2

```
1  CREATE TABLE TB_GRADUATE_CANDIDATE (
2      Student_ID      NUMBER(6) PRIMARY KEY,
3      Last_Name       VARCHAR2(50) NOT NULL,
4      First_Name      VARCHAR2(50) NOT NULL,
5      Credits         NUMBER(3) NOT NULL,
6      Graduation_Date DATE NOT NULL
7  );
```

Results | Explain | Describe | Saved SQL | History

Table created.

0.07 seconds

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```
1  desc TB_GRADUATE_CANDIDATE
```

Results | Explain | **Describe** | Saved SQL | History

Object Type **TABLE** Object **TB_GRADUATE_CANDIDATE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TB_GRADUATE_CANDIDATE	STUDENT_ID	NUMBER	-	6	0	1	-	-	-
	LAST_NAME	VARCHAR2	50	-	-	-	-	-	-
	FIRST_NAME	VARCHAR2	50	-	-	-	-	-	-
	CREDITS	NUMBER	-	3	0	-	-	-	-
	GRADUATION_DATE	DATE	7	-	-	-	-	-	-

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Laboratory Activity #3

```
1   CREATE TABLE TB_FACULTY (
2     Faculty_ID      NUMBER(5) PRIMARY KEY,
3     Last_Name       VARCHAR2(50) NOT NULL,
4     First_Name      VARCHAR2(50) NOT NULL,
5     Title           VARCHAR(50) NOT NULL,
6     Emale           VARCHAR(100) NOT NULL,
7     Phone           NUMBER(4) NOT NULL,
8     Offce           VARCHAR(30) NOT NULL
9   );
```

Results | Explain | Describe | Saved SQL | History

Table created.

0.08 seconds

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```
1   desc TB_FACULTY
```

Results | Explain | **Describe** | Saved SQL | History

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TB_FACULTY	FACULTY_ID	NUMBER	-	5	0	1	-	-	-
	LAST_NAME	VARCHAR2	50	-	-	-	-	-	-
	FIRST_NAME	VARCHAR2	50	-	-	-	-	-	-
	TITLE	VARCHAR2	50	-	-	-	-	-	-
	EMALE	VARCHAR2	100	-	-	-	-	-	-
	PHONE	NUMBER	-	4	0	-	-	-	-
	OFFCE	VARCHAR2	30	-	-	-	-	-	-

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Laboratory Activity #4

```
1   CREATE TABLE TB_DEPARTMENT (
2     Dept_ID      NUMBER(6) PRIMARY KEY,
3     Dept_Name       VARCHAR2(50) NOT NULL,
4     Location      VARCHAR2(50) NOT NULL
5   );
```

Results | Explain | Describe | Saved SQL | History

```
Table created.

0.04 seconds
```

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```
1   desc TB_DEPARTMENT
2
```

Results | Explain | **Describe** | Saved SQL | History

Object Type **TABLE** Object **TB_DEPARTMENT**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TB_DEPARTMENT	DEPT_ID	NUMBER	-	6	0	1	-	-	-
	DEPT_NAME	VARCHAR2	50	-	-	-	-	-	-
	LOCATION	VARCHAR2	50	-	-	-	-	-	-

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```
1  CREATE TABLE TB_EMPLOYEE (
2    Emp_ID       NUMBER(6) PRIMARY KEY,
3    Emp_FirstName VARCHAR2(50) NOT NULL,
4    Emp_Lastname  VARCHAR2(50) NOT NULL,
5    Emp_HireDate  DATE NOT NULL,
6    Salary        NUMBER(7,2) NOT NULL,
7    Dept_ID       NUMBER(6) REFERENCES TB_DEPARTMENT(Dept_ID)
8  );
```

Results | Explain | Describe | Saved SQL | History

Table created.

0.08 seconds

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```
1  desc TB_EMPLOYEE
```

Results | Explain | Describe | Saved SQL | History

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TB_EMPLOYEE	EMP_ID	NUMBER	-	6	0	1	-	-	-
	EMP_FIRSTNAME	VARCHAR2	50	-	-	-	-	-	-
	EMP_LASTNAME	VARCHAR2	50	-	-	-	-	-	-
	EMP_HIREDATE	DATE	7	-	-	-	-	-	-
	SALARY	NUMBER	-	7	2	-	-	-	-
	DEPT_ID	NUMBER	-	6	0	-	✓	-	-

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Laboratory Activity #5

```
1  ALTER TABLE TB_EMPLOYEE ADD Position VARCHAR2(30);
2  ALTER TABLE TB_EMPLOYEE MODIFY Emp_Lastname VARCHAR2(60);
3
4  desc TB_EMPLOYEE
5
```

Results | Explain | **Describe** | Saved SQL | History

Object Type **TABLE** Object **TB_EMPLOYEE**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TB_EMPLOYEE	EMP_ID	NUMBER	-	6	0	1	-	-	-
	EMP_FIRSTNAME	VARCHAR2	50	-	-	-	-	-	-
	EMP_LASTNAME	VARCHAR2	60	-	-	-	-	-	-
	EMP_HIREDATE	DATE	7	-	-	-	-	-	-
	SALARY	NUMBER	-	7	2	-	-	-	-
	DEPT_ID	NUMBER	-	6	0	-	✓	-	-
	POSITION	VARCHAR2	30	-	-	-	✓	-	-

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```
1
2  DROP TABLE TB_EMPLOYEE
3  desc TB_EMPLOYEE
```

Results | Explain | **Describe** | Saved SQL | History

Object to be described could not be found.

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Laboratory Activity #6

```
INSERT INTO TB_GRADUATE_CANDIDATE VALUES
  (111111, 'Abletes',  'Julianne',   154, TO_DATE('04/01/19','MM/DD/RR')),
  (222222, 'Bibat',    'Archimedes', 174, TO_DATE('04/01/19','MM/DD/RR')),
  (333333, 'Bongolan', 'Glenn',      150, TO_DATE('04/01/19','MM/DD/RR')),
  (444444, 'Carluen',  'Ramon',      200, TO_DATE('04/01/19','MM/DD/RR')),
  (555555, 'Dada',     'Mateo',      174, TO_DATE('04/01/19','MM/DD/RR')),
  (666666, 'Mendoza',  'Marianne',   154, TO_DATE('04/01/19','MM/DD/RR')),
  (777777, 'Silvino',  'Jenny',      200, TO_DATE('04/01/19','MM/DD/RR')),
  (888888, 'Tamayo',   'Ronaldo',    200, TO_DATE('04/01/19','MM/DD/RR')),
  (999999, 'Taquiqui', 'Cathy',      154, TO_DATE('04/01/19','MM/DD/RR')),
  (111110, 'Ulanday',  'Yuan',       174, TO_DATE('04/01/19','MM/DD/RR'));
```

Results | Explain | Describe | Saved SQL | History

10 row(s) inserted.

0.06 seconds

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```
SELECT * FROM TB_GRADUATE_CANDIDATE;
```

Results | Explain | Describe | Saved SQL | History

STUDENT_ID	LAST_NAME	FIRST_NAME	CREDITS	GRADUATIO
111111	Abletes	Julianne	154	4/1/2019
222222	Bibat	Archimedes	174	4/1/2019
333333	Bongolan	Glenn	150	4/1/2019
444444	Carluen	Ramon	200	4/1/2019
555555	Dada	Mateo	174	4/1/2019
666666	Mendoza	Marianne	154	4/1/2019
777777	Silvino	Jenny	200	4/1/2019
888888	Tamayo	Ronaldo	200	4/1/2019
999999	Taquiqui	Cathy	154	4/1/2019
111110	Ulanday	Yuan	174	4/1/2019

10 rows returned in 0.15 seconds Download

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Laboratory Activity #7

```
INSERT INTO TB_FACULTY VALUES
  (11110, 'Delas Alas', 'Rhiana',      'Visiting Professor',  'rdelasalas@gmail.com',    3017, 'Bldg G'),
  (11111, 'De Vera',    'Reynaldo',    'Full Time Professor', 'rdevera@gmail.com',       4201, 'Bldg H'),
  (11112, 'Trinidad',   'Renbert',     'Part Time Professor', 'rtrinidad@yahoo.com',     5210, 'Bldg C'),
  (11113, 'De Guzman',  'James',       'Visiting Professor',  'jdeguzman@yahoo.com',     5211, 'Bldg H'),
  (11114, 'Perey',      'Joel',        'Research Professor',  'jperey@yahoo.com',        3012, 'Bldg A'),
  (11115, 'Ponciano',   'Mary Ann',    'Full Time Professor', 'mponciano@yahoo.com',     2011, 'Bldg H'),
  (11116, 'Marges',     'Lailani',     'Full Time Professor', 'lmarge@yahoo.com',        1001, 'Bldg C'),
  (11117, 'Redondon',   'Yumi',        'Visiting Professor',  'yredondon@gmail.com',     2531, 'Bldg C'),
  (11118, 'Ramos',      'Ross Jericho','Part Time Professor', 'rjramos@gmail.com',       2011, 'Bldg A'),
  (11119, 'Mararagan',  'Sean Eriq',   'Full Time Professor', 'semararagan@yahoo.com',   2031, 'Bldg G');

SELECT * FROM TB_FACULTY;
```

Results | Explain | Describe | Saved SQL | History

FACULTY_ID	LAST_NAME	FIRST_NAME	TITLE	EMALE	PHONE	OFFCE
11110	Delas Alas	Rhiana	Visiting Professor	rdelasalas@gmail.com	3017	Bldg G
11111	De Vera	Reynaldo	Full Time Professor	rdevera@gmail.com	4201	Bldg H
11112	Trinidad	Renbert	Part Time Professor	rtrinidad@yahoo.com	5210	Bldg C
11113	De Guzman	James	Visiting Professor	jdeguzman@yahoo.com	5211	Bldg H
11114	Perey	Joel	Research Professor	jperey@yahoo.com	3012	Bldg A
11115	Ponciano	Mary Ann	Full Time Professor	mponciano@yahoo.com	2011	Bldg H
11116	Marges	Lailani	Full Time Professor	lmarge@yahoo.com	1001	Bldg C
11117	Redondon	Yumi	Visiting Professor	yredondon@gmail.com	2531	Bldg C
11118	Ramos	Ross Jericho	Part Time Professor	rjramos@gmail.com	2011	Bldg A
11119	Mararagan	Sean Eriq	Full Time Professor	semararagan@yahoo.com	2031	Bldg G

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```
1   UPDATE TB_FACULTY
2   SET Title = 'Full Time Professor'
3   WHERE Faculty_ID = 11118;
4
5   UPDATE TB_FACULTY
6   SET Email = 'reynaldo.devera@gmail.com'
7   WHERE Faculty_ID = 11111;
8
9   DELETE FROM TB_FACULTY
10  WHERE Faculty_ID = 11117;
11
12  SELECT * FROM TB_FACULTY;
```

Results | Explain | Describe | Saved SQL | History

FACULTY_ID	LAST_NAME	FIRST_NAME	TITLE	EMAIL	PHONE	OFFCE
11110	Delas Alas	Rhiana	Visiting Professor	rdelasalas@gmail.com	3017	Bldg G
11111	De Vera	Reynaldo	Full Time Professor	reynaldo.devera@gmail.com	4201	Bldg H
11112	Trinidad	Renbert	Part Time Professor	rtrinidad@yahoo.com	5210	Bldg C
11113	De Guzman	James	Visiting Professor	jdeguzman@yahoo.com	5211	Bldg H
11114	Perey	Joel	Research Professor	jperey@yahoo.com	3012	Bldg A
11115	Ponciano	Mary Ann	Full Time Professor	mponciano@yahoo.com	2011	Bldg H
11116	Marges	Lailani	Full Time Professor	lmarge@yahoo.com	1001	Bldg C
11118	Ramos	Ross Jericho	Full Time Professor	rjramos@gmail.com	2011	Bldg A
11119	Mararagan	Sean Eriq	Full Time Professor	semararagan@yahoo.com	2031	Bldg G

9 rows returned in 0.01 seconds Download

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Laboratory Activity #8

```
1  INSERT INTO TB_DEPARTMENT VALUES
2    (999990, 'Accounting', '3F Rm 301'),
3    (888880, 'Marketing',  '4F Rm 411'),
4    (777770, 'Purchasing', '4F Rm 413'),
5    (666660, 'Sales',      '3F Rm 312'),
6    (555550, 'Payroll',    '2F Rm 201');
7
8    SELECT * FROM TB_DEPARTMENT;
```

Results | Explain | Describe | Saved SQL | History

DEPT_ID	DEPT_NAME	LOCATION
999990	Accounting	3F Rm 301
888880	Marketing	4F Rm 411
777770	Purchasing	4F Rm 413
666660	Sales	3F Rm 312
555550	Payroll	2F Rm 201

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```
INSERT INTO TB_EMPLOYEE VALUES
  (222220, 'Rashley',   'Torres',      'Manager',    TO_DATE('02/25/1999','MM/DD/YYYY'), 53000, 999990),
  (222221, 'Joanna',    'Mayuga',      'Manager',    TO_DATE('03/15/1996','MM/DD/YYYY'), 54000, 888880),
  (222222, 'Martha',    'Rosas',       'Manager',    TO_DATE('05/30/1997','MM/DD/YYYY'), 53000, 777770),
  (222223, 'Marissa',   'Abad',        'Manager',    TO_DATE('02/25/1999','MM/DD/YYYY'), 53000, 666660),
  (222224, 'Marris',    'Del Rosario', 'Analyst',    TO_DATE('09/11/1999','MM/DD/YYYY'), 53000, 999990),
  (222225, 'Alex',      'Causaren',    'Clerk',      TO_DATE('08/02/1993','MM/DD/YYYY'), 54000, 999990),
  (222226, 'Alley',     'Reyes',       'Manager',    TO_DATE('04/03/1994','MM/DD/YYYY'), 55000, 555550),
  (222227, 'Aaron',     'Sabater',     'Salesman',   TO_DATE('03/12/2000','MM/DD/YYYY'), 25500, 999990),
  (222228, 'Joseph',    'Trinidad',    'Clerk',      TO_DATE('07/26/2001','MM/DD/YYYY'), 19000, 777770),
  (222229, 'Christian', 'Delo Santos', 'Salesman',   TO_DATE('10/23/2005','MM/DD/YYYY'), 27000, 777770),
  (222230, 'Jeffrey',   'Romero',      'Clerk',      TO_DATE('10/26/2000','MM/DD/YYYY'), 18500, 888880),
  (222231, 'Martheena', 'Herrera',     'Salesman',   TO_DATE('11/22/1999','MM/DD/YYYY'), 30000, 888880),
  (222232, 'Martin',    'Mesias',      'Analyst',    TO_DATE('09/11/2000','MM/DD/YYYY'), 48000, 888880),
  (222233, 'Arnie',     'Agtonton',    'Analyst',    TO_DATE('05/21/1994','MM/DD/YYYY'), 50000, 777770),
  (222234, 'Mark',      'Redublo',     'Salesman',   TO_DATE('04/16/2001','MM/DD/YYYY'), 30000, 666660),
  (222235, 'Valentino', 'Saez',        'Supervisor', TO_DATE('12/19/2000','MM/DD/YYYY'), 45000, 555550);

SELECT * FROM TB_EMPLOYEE;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	POSITION	EMP_HIREDATE	SALARY	DEPT_ID
222220	Rashley	Torres	Manager	2/25/1999	53000	999990
222221	Joanna	Mayuga	Manager	3/15/1996	54000	888880
222222	Martha	Rosas	Manager	5/30/1997	53000	777770
222223	Marissa	Abad	Manager	2/25/1999	53000	666660
222224	Marris	Del Rosario	Analyst	9/11/1999	53000	999990
222225	Alex	Causaren	Clerk	8/2/1993	54000	999990
222226	Alley	Reyes	Manager	4/3/1994	55000	555550
222227	Aaron	Sabater	Salesman	3/12/2000	25500	999990
222228	Joseph	Trinidad	Clerk	7/26/2001	19000	777770
222229	Christian	Delo Santos	Salesman	10/23/2005	27000	777770

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Laboratory Activity #10

1. **Create table TB_DOCSTATUS with the following fields and attributes:**
2. **Confirm creation of the table using DESCRIBE statement.**

```
1  CREATE TABLE TB_DOCSTATUS (
2    DOC_ID   VARCHAR2(4)  PRIMARY KEY,
3    COMPANY  VARCHAR2(50) NOT NULL
4  );
5
6  desc TB_DOCSTATUS;
```

Results | Explain | **Describe** | Saved SQL | History

Object Type **TABLE** Object **TB_DOCSTATUS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TB_DOCSTATUS	DOC_ID	VARCHAR2	4	-	-	1	-	-	-
	COMPANY	VARCHAR2	50	-	-	-	-	-	-

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3. Create table TB_STATUS with the following fields and attributes:

```
1  CREATE TABLE TB_STATUS (
2    STATUS       VARCHAR2(1)  PRIMARY KEY,
3    SDESCRIPTION VARCHAR2(10) NOT NULL
4  );
5
6  desc TB_STATUS;
```

Results | Explain | **Describe** | Saved SQL | History

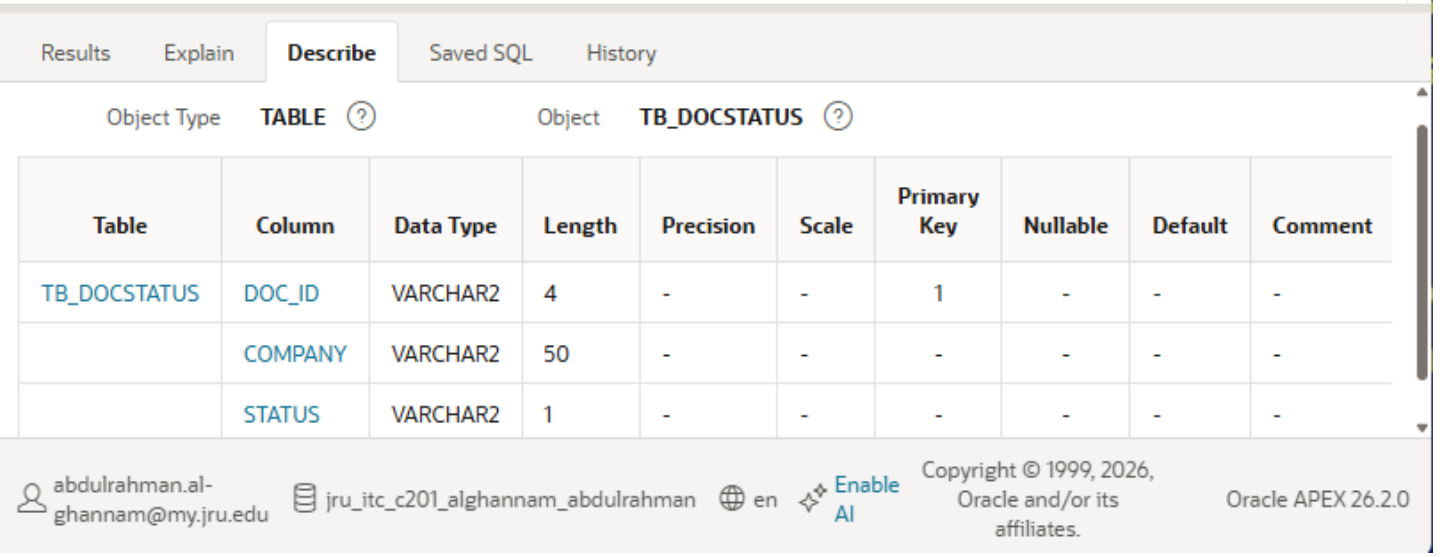
Object Type **TABLE** Object **TB_STATUS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TB_STATUS	STATUS	VARCHAR2	1	-	-	1	-	-	-
	SDESCRIPTION	VARCHAR2	10	-	-	-	-	-	-

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4. Add the field to the TB_DOCSTATUS table:

```
1   ALTER TABLE TB_DOCSTATUS ADD STATUS VARCHAR2(1) NOT NULL;
2
3   desc TB_DOCSTATUS;
```



Results | Explain | **Describe** | Saved SQL | History

Object Type **TABLE** Object **TB_DOCSTATUS**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
TB_DOCSTATUS	DOC_ID	VARCHAR2	4	-	-	1	-	-	-
	COMPANY	VARCHAR2	50	-	-	-	-	-	-
	STATUS	VARCHAR2	1	-	-	-	-	-	-

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5. Add the following records in the TB_DOCSTATUS table:

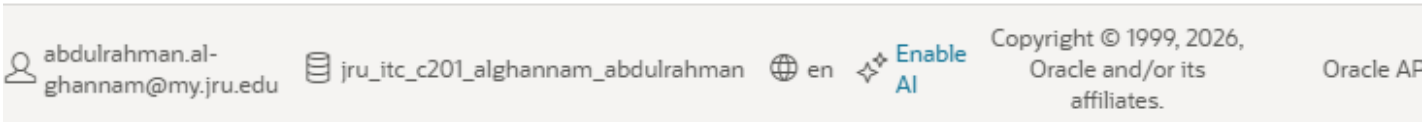
6. Confirm creation of the table using DESCRIBE statement.

```
1   INSERT INTO TB_DOCSTATUS VALUES ('2001', 'Teewai.com Inc', 'I');
2   INSERT INTO TB_DOCSTATUS VALUES ('2002', 'Fave Company', 'A');
3
4   select * from TB_DOCSTATUS;
```

Results | Explain | Describe | Saved SQL | History

DOC_ID	COMPANY	STATUS
2001	Teewai.com Inc	I
2002	Fave Company	A

2 rows returned in 0.02 seconds Download



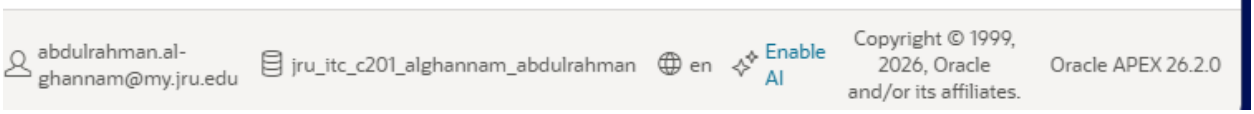
7. Add the following records in the TB_STATUS table:

```
1   INSERT INTO TB_STATUS VALUES ('I', 'Inactive');
2   INSERT INTO TB_STATUS VALUES ('A', 'Active');
3   INSERT INTO TB_STATUS VALUES ('S', 'Suspended');
4
5
6   select * from TB_STATUS;
```

Results | Explain | Describe | Saved SQL | History

STATUS	SDESCRIPTION
I	Inactive
A	Active
S	Suspended

3 rows returned in 0.02 seconds Download



8. Change the status of Teewai.com Inc. to S.

```
1   UPDATE TB_DOCSTATUS SET STATUS = 'S'
2   WHERE COMPANY = 'Teewai.com Inc';
3
4   select * from TB_DOCSTATUS;
```

Results | Explain | Describe | Saved SQL | History

DOC_ID	COMPANY	STATUS
2001	Teewai.com Inc	S
2002	Fave Company	A

2 rows returned in 0.01 seconds Download

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9. Delete the record in the TB_STATUS table with the Status I.

```
1   DELETE FROM TB_STATUS WHERE STATUS = 'I';
2
3   select * from TB_STATUS;
```

Results | Explain | Describe | Saved SQL | History

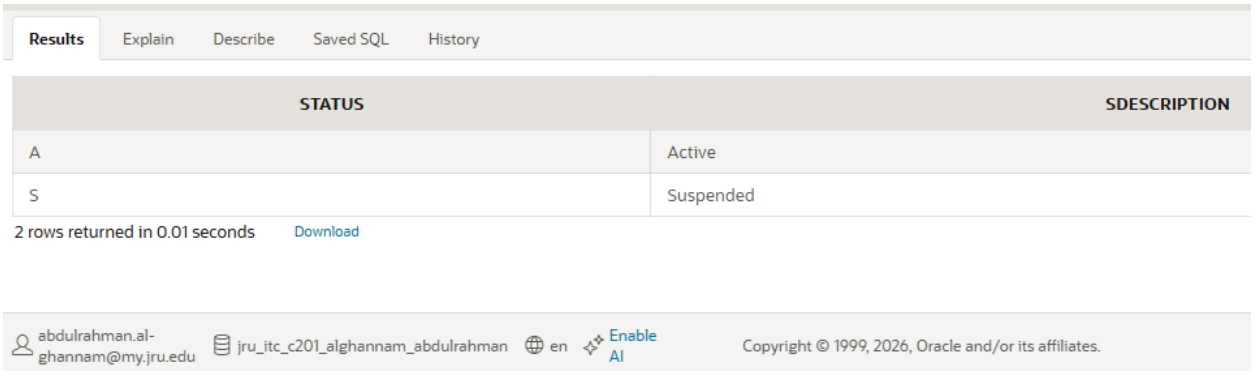
STATUS	SDESCRIPTION
A	Active
S	Suspended

2 rows returned in 0.00 seconds Download

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10. Display all records in the TB_STATUS table.

```
1  SELECT * FROM TB_STATUS;
2  SELECT * FROM TB_DOCSTATUS;
```



Results | Explain | Describe | Saved SQL | History

STATUS	SDESCRIPTION
A	Active
S	Suspended

2 rows returned in 0.01 seconds Download

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11. Display all records in the TB_DOCSTATUS table.

```
1  SELECT * FROM TB_STATUS;
2  SELECT * FROM TB_DOCSTATUS;
```



Results | Explain | Describe | Saved SQL | History

DOC_ID	COMPANY	STATUS
2001	Teewai.com Inc	S
2002	Fave Company	A

2 rows returned in 0.01 seconds Download

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Laboratory Activity #11

1. Add the following fields in your TB_DOCSTATUS table:

2. Add the following records in the TB_DOCSTATUS table:

3. Display all records with null Capital or TermExist fields.

```
ALTER TABLE TB_DOCSTATUS ADD CAPITAL NUMBER(9,2);
ALTER TABLE TB_DOCSTATUS ADD TERMEXIST NUMBER(3);

INSERT INTO TB_DOCSTATUS VALUES ('2003', 'Goldilocks Bakeshop Corp', 'A', 500000, 100);
INSERT INTO TB_DOCSTATUS VALUES ('2004', 'Collegian Exponent Inc.', 'S', 100000, 50);
INSERT INTO TB_DOCSTATUS VALUES ('2005', 'Uniphil Computer Inc', 'A', NULL, 50);

select * from TB_DOCSTATUS
```

Results | Explain | Describe | Saved SQL | History

DOC_ID	COMPANY	STATUS	CAPITAL	TERMEXIST
2001	Teewai.com Inc	S	-	-
2002	Fave Company	A	-	-
2003	Goldilocks Bakeshop Corp	A	500000	100
2005	Uniphil Computer Inc	A	-	50
2004	Collegian Exponent Inc.	S	100000	50

5 rows returned in 0.02 seconds Download

4. Change all null Capital values to 250000.

5. Change all null TermExist values to 250

```
UPDATE TB_DOCSTATUS SET CAPITAL = 250000 WHERE CAPITAL IS NULL;

UPDATE TB_DOCSTATUS SET TERMEXIST = 250 WHERE TERMEXIST IS NULL;

select * from TB_DOCSTATUS
```

Results | Explain | Describe | Saved SQL | History

DOC_ID	COMPANY	STATUS	CAPITAL	TERME
2001	Teewai.com Inc	S	250000	250
2002	Fave Company	A	250000	250
2003	Goldilocks Bakeshop Corp	A	500000	100
2005	Uniphil Computer Inc	A	250000	50
2004	Collegian Exponent Inc.	S	100000	50

5 rows returned in 0.01 seconds Download

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6. Change the Capital value of Uniphil Computer to 300000.

```
1  UPDATE TB_DOCSTATUS SET CAPITAL = 300000
2  WHERE DOC_ID = '2005';
3
4  select * from TB_DOCSTATUS
```

Results | Explain | Describe | Saved SQL | History

DOC_ID	COMPANY	STATUS	CAPITAL	TERMEXIST
2001	Teewai.com Inc	S	250000	250
2002	Fave Company	A	250000	250
2003	Goldilocks Bakeshop Corp	A	500000	100
2005	Uniphil Computer Inc	A	300000	50
2004	Collegian Exponent Inc.	S	100000	50

5 rows returned in 0.01 seconds Download

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7. From the DocStatus table, display all active companies without O or U in the name.

```
1  SELECT *
2  FROM TB_DOCSTATUS
3  WHERE STATUS = 'A'
4    AND COMPANY NOT LIKE ('%O%')
5    AND COMPANY NOT LIKE ('%U%');
```

Results | Explain | Describe | Saved SQL | History

DOC_ID	COMPANY	STATUS	CAPITAL	TERMEXIST
2002	Fave Company	A	250000	250
2003	Goldilocks Bakeshop Corp	A	500000	100

2 rows returned in 0.01 seconds Download

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8. Display the Doc_ID, Company name and the Capital of suspended companies with more than 200000 Capital

```
1   SELECT DOC_ID, COMPANY, CAPITAL FROM TB_DOCSTATUS
2   WHERE STATUS = 'S' AND CAPITAL > 200000;
```

Results | Explain | Describe | Saved SQL | History

DOC_ID	COMPANY	CAPITAL
2001	Teewai.com Inc	250000

1 rows returned in 0.02 seconds Download

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9. Display only the distinct TB_STATUS field values.

```
1   SELECT DISTINCT STATUS FROM TB_DOCSTATUS;
```

Results | Explain | Describe | Saved SQL | History

STATUS
S
A

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10. Display the two largest Capital and their corresponding Company names in the TB_DOCSTATUS table.

```
1   SELECT COMPANY, CAPITAL FROM TB_DOCSTATUS
2   ORDER BY CAPITAL DESC
3   FETCH FIRST 2 ROWS ONLY;
```

Results | Explain | Describe | Saved SQL | History

COMPANY	CAPITAL
Goldilocks Bakeshop Corp	500000
Uniphil Computer Inc	300000

2 rows returned in 0.02 seconds Download

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11. Display the average and sum of Capital with column headings Average and Total, respectively.

```
1   SELECT AVG(CAPITAL) AS "Average", SUM(CAPITAL) AS "Total"
2   FROM TB_DOCSTATUS;
```

Results | Explain | Describe | Saved SQL | History

Average	Total
280000	1400000

1 rows returned in 0.01 seconds Download

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12. Display the minimum and maximum of TermExist with column headings Minimum and Maximum, respectively.

```
1  SELECT MIN(TERMEXIST) AS "Minimum", MAX(TERMEXIST) AS "Maximum"
2  FROM TB_DOCSTATUS;
```

Results | Explain | Describe | Saved SQL | History

Minimum	Maximum
50	250

1 rows returned in 0.00 seconds Download

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13. Using an aggregate function, get the total number of companies with 50 years of term of existence.

```
1  SELECT COUNT(*) FROM TB_DOCSTATUS WHERE TERMEXIST = 50;
```

Results | Explain | Describe | Saved SQL | History

COUNT(*)
2

1 rows returned in 0.00 seconds Download

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14. Add the following records in the TB_STATUS table:

```
1   INSERT INTO TB_STATUS VALUES ('W', 'Widthdrawn');
2   INSERT INTO TB_STATUS VALUES ('O', 'Abandones');
3
4   select * from TB_STATUS
```

Results | Explain | Describe | Saved SQL | History

STATUS	SDESCRIPTION
W	Widthdrawn
A	Active
O	Abandones
S	Suspended

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15. From the Status table, display all non-active Status.

```
1   SELECT * FROM TB_STATUS WHERE STATUS <> 'A';
```

Results | Explain | Describe | Saved SQL | History

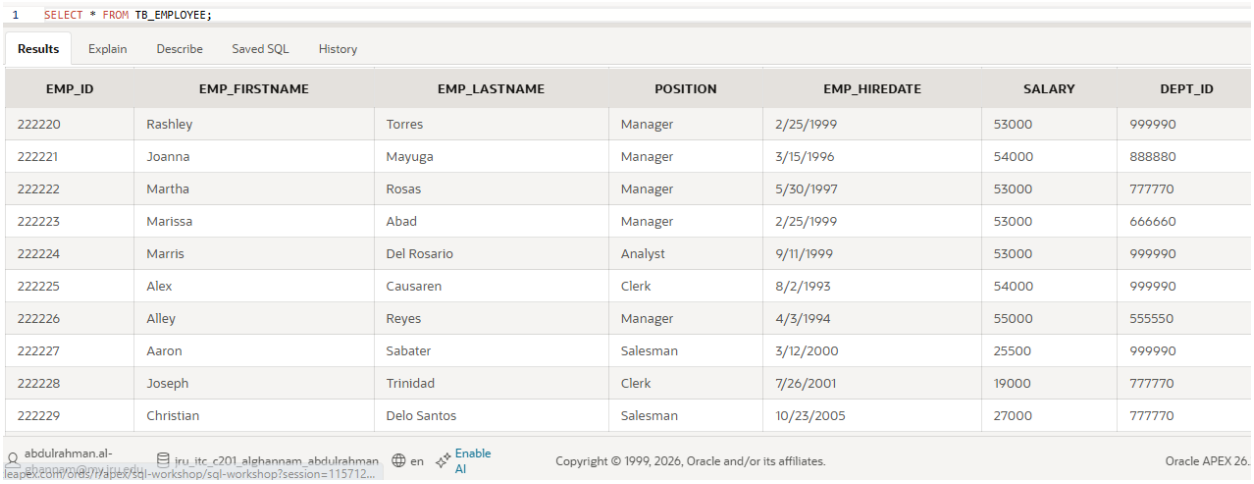
STATUS	SDESCRIPTION
W	Widthdrawn
O	Abandones
S	Suspended

3 rows returned in 0.01 seconds Download

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Laboratory Activity #12

1. Open the exiting TB_EMPLOYEE table.



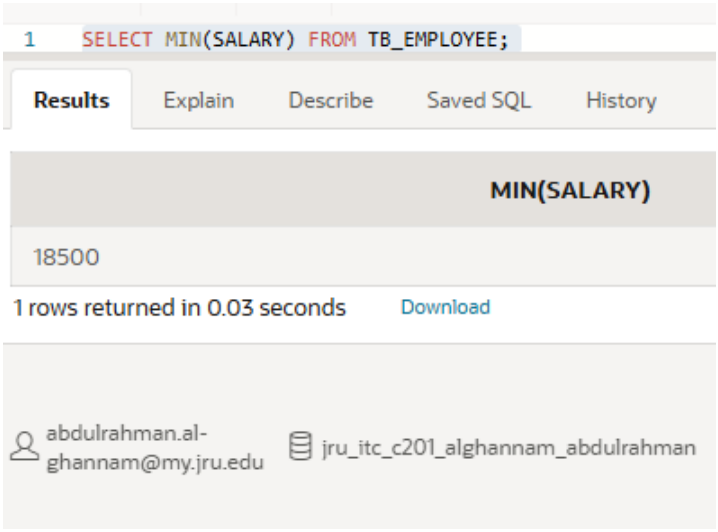
```
1  SELECT * FROM TB_EMPLOYEE;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	POSITION	EMP_HIREDATE	SALARY	DEPT_ID
222220	Rashley	Torres	Manager	2/25/1999	53000	999990
222221	Joanna	Mayuga	Manager	3/15/1996	54000	888880
222222	Martha	Rosas	Manager	5/30/1997	53000	777770
222223	Marissa	Abad	Manager	2/25/1999	53000	666660
222224	Marris	Del Rosario	Analyst	9/11/1999	53000	999990
222225	Alex	Causaren	Clerk	8/2/1993	54000	999990
222226	Alley	Reyes	Manager	4/3/1994	55000	555550
222227	Aaron	Sabater	Salesman	3/12/2000	25500	999990
222228	Joseph	Trinidad	Clerk	7/26/2001	19000	777770
222229	Christian	Delo Santos	Salesman	10/23/2005	27000	777770

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2. Get the minimum salary in the TB_EMPLOYEE table.



```
1  SELECT MIN(SALARY) FROM TB_EMPLOYEE;
```

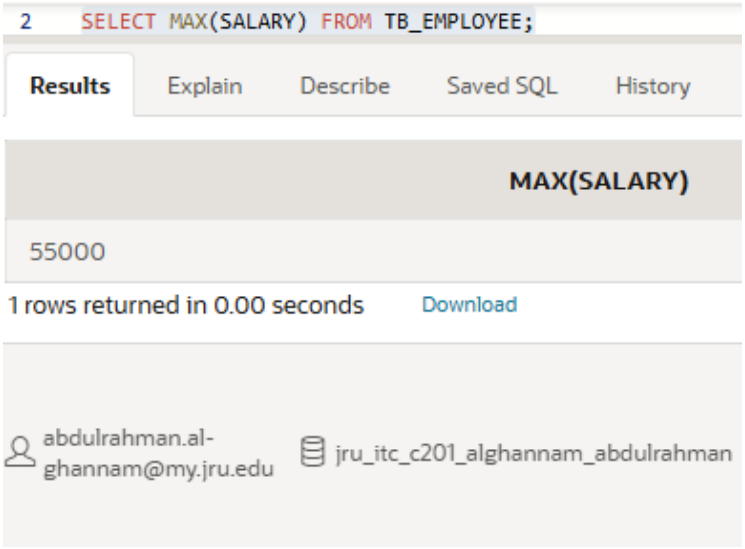
Results | Explain | Describe | Saved SQL | History

MIN(SALARY)
18500

1 rows returned in 0.03 seconds Download

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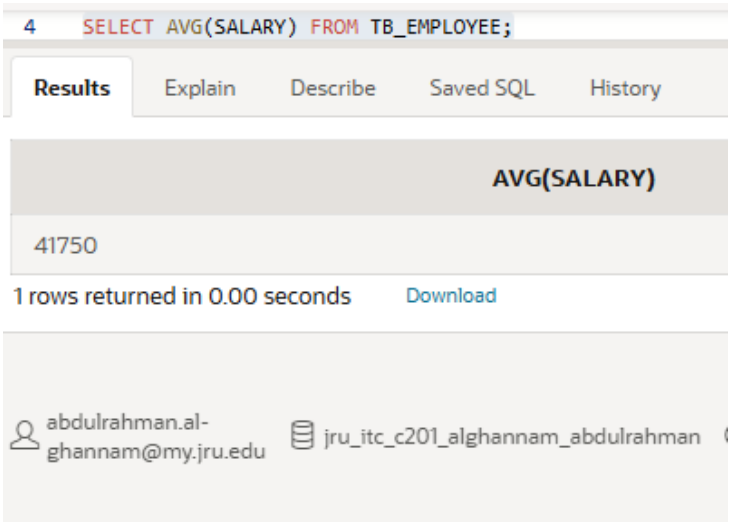
3. Get the maximum salary in the TB_EMPLOYEE table.



4. Get the total salary in the TB_EMPLOYEE table.



5. Get the average salary in the TB_EMPLOYEE table.



7. Give the results of this query:

SELECT EMP_FIRSTNAME, EMP_LASTNAME, DEPT_NAME

FROM TB_EMPLOYEE NATURAL JOIN TB_DEPARTMENT

```
1  SELECT EMP_FIRSTNAME, EMP_LASTNAME, DEPT_NAME
2  FROM TB_EMPLOYEE NATURAL JOIN TB_DEPARTMENT
```

Results | Explain | Describe | Saved SQL | History

EMP_FIRSTNAME	EMP_LASTNAME	DEPT_NAME
Valentino	Saez	Payroll
Alley	Reyes	Payroll
Mark	Redublo	Sales
Marissa	Abad	Sales
Martha	Rosas	Purchasing
Arnie	Agtonton	Purchasing
Christian	Delo Santos	Purchasing
Joseph	Trinidad	Purchasing
Joanna	Mayuga	Marketing
Martin	Mesias	Marketing

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8. Give the results of this query:

SELECT EMP_FIRSTNAME, EMP_LASTNAME, POSITION, DEPT_ID,

DEPT_NAME

FROM TB_EMPLOYEE JOIN TB_DEPARTMENT USING (DEPT_ID)

```
1  SELECT EMP_FIRSTNAME, EMP_LASTNAME, POSITION, DEPT_ID,
2  DEPT_NAME
3  FROM TB_EMPLOYEE JOIN TB_DEPARTMENT USING (DEPT_ID)
```

Results | Explain | Describe | Saved SQL | History

EMP_FIRSTNAME	EMP_LASTNAME	POSITION	DEPT_ID	DEPT_NAME
Valentino	Saez	Supervisor	555550	Payroll
Alley	Reyes	Manager	555550	Payroll
Mark	Redublo	Salesman	666660	Sales
Marissa	Abad	Manager	666660	Sales
Martha	Rosas	Manager	777770	Purchasing
Arnie	Agtonton	Analyst	777770	Purchasing
Christian	Delo Santos	Salesman	777770	Purchasing
Joseph	Trinidad	Clerk	777770	Purchasing
Joanna	Mayuga	Manager	888880	Marketing
Martin	Mesias	Analyst	888880	Marketing

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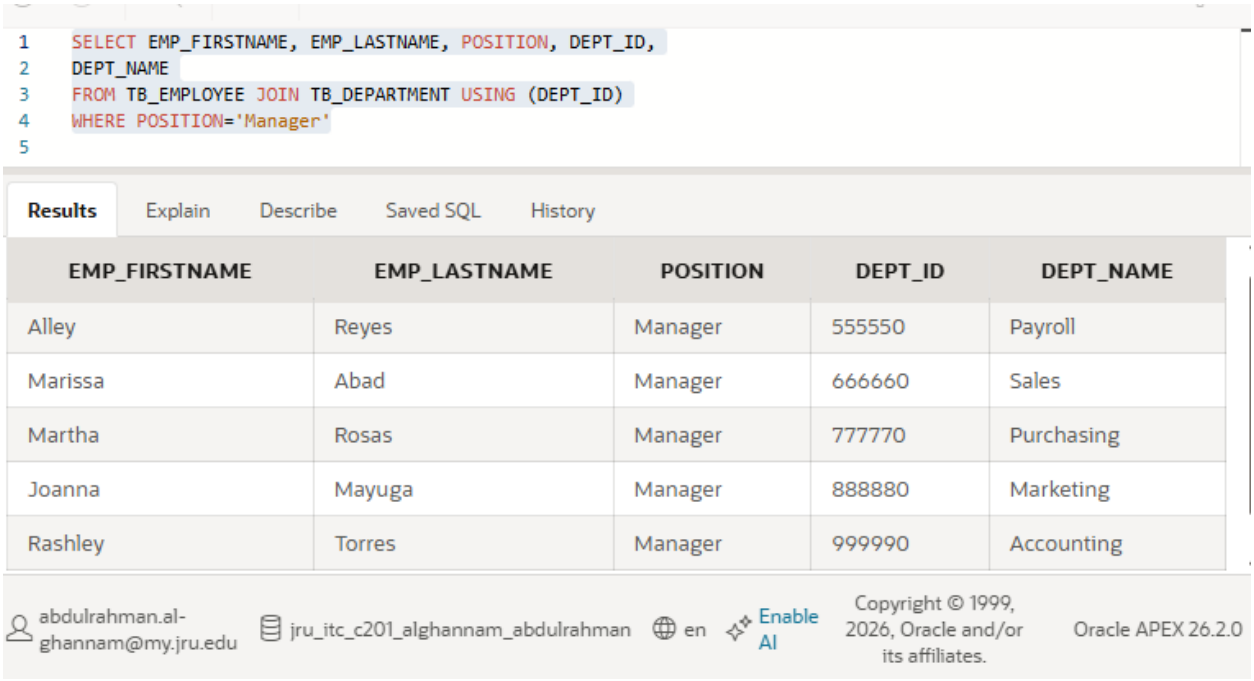
9. Give the results of this query:

SELECT EMP_FIRSTNAME, EMP_LASTNAME, POSITION, DEPT_ID,

DEPT_NAME

FROM TB_EMPLOYEE JOIN TB_DEPARTMENT USING (DEPT_ID)

WHERE POSITION='MANAGER'



```
1  SELECT EMP_FIRSTNAME, EMP_LASTNAME, POSITION, DEPT_ID,
2  DEPT_NAME
3  FROM TB_EMPLOYEE JOIN TB_DEPARTMENT USING (DEPT_ID)
4  WHERE POSITION='Manager'
5
```

Results | Explain | Describe | Saved SQL | History

EMP_FIRSTNAME	EMP_LASTNAME	POSITION	DEPT_ID	DEPT_NAME
Alley	Reyes	Manager	555550	Payroll
Marissa	Abad	Manager	666660	Sales
Martha	Rosas	Manager	777770	Purchasing
Joanna	Mayuga	Manager	888880	Marketing
Rashley	Torres	Manager	999990	Accounting

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10. Give the results of this query:

SELECT EMP_FIRSTNAME, EMP_LASTNAME, DEPT_NAME

FROM TB_EMPLOYEE E JOIN TB_DEPARTMENT D

ON (E.DEPT_ID=D.DEPT_ID)

```
1  SELECT EMP_FIRSTNAME, EMP_LASTNAME, DEPT_NAME
2  FROM TB_EMPLOYEE E JOIN TB_DEPARTMENT D
3  ON (E.DEPT_ID=D.DEPT_ID)
```

Results | Explain | Describe | Saved SQL | History

EMP_FIRSTNAME	EMP_LASTNAME	DEPT_NAME
Valentino	Saez	Payroll
Alley	Reyes	Payroll
Mark	Redublo	Sales
Marissa	Abad	Sales
Martha	Rosas	Purchasing
Arnie	Agtonton	Purchasing
Christian	Delo Santos	Purchasing
Joseph	Trinidad	Purchasing
Joanna	Mayuga	Marketing
Martin	Mesias	Marketing

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11. Give the results of this query:

SELECT EMP_FIRSTNAME, EMP_LASTNAME, DEPT_NAME

FROM TB_EMPLOYEE E JOIN TB_DEPARTMENT D

ON (E.DEPT_ID=D.DEPT_ID)

WHERE EMP_LASTNAME LIKE '%M%'

```
1  SELECT EMP_FIRSTNAME, EMP_LASTNAME, DEPT_NAME
2  FROM TB_EMPLOYEE E JOIN TB_DEPARTMENT D
3  ON (E.DEPT_ID=D.DEPT_ID)
```

Results | Explain | Describe | Saved SQL | History

EMP_FIRSTNAME	EMP_LASTNAME	DEPT_NAME
Joanna	Mayuga	Marketing
Martin	Mesias	Marketing

2 rows returned in 0.01 seconds Download

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Problem 1. INNER JOIN

Display the employee ID, first name, last name, position, department name, and department location of employees who have a matching department.

```
select e.EMP_ID , e.EMP_FIRSTNAME, e.EMP_LASTNAME , e.POSITION , d.DEPT_NAME , d.LOCATION
from TB_EMPLOYEE e inner join TB_DEPARTMENT d
on (e.DEPT_ID = d.DEPT_ID);
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	POSITION	DEPT_NAME	LOCATION
222235	Valentino	Saez	Supervisor	Payroll	2F Rm 201
222226	Alley	Reyes	Manager	Payroll	2F Rm 201
222234	Mark	Redublo	Salesman	Sales	3F Rm 312
222223	Marissa	Abad	Manager	Sales	3F Rm 312
222222	Martha	Rosas	Manager	Purchasing	4F Rm 413
222233	Arnie	Agtonton	Analyst	Purchasing	4F Rm 413
222229	Christian	Delo Santos	Salesman	Purchasing	4F Rm 413
222228	Joseph	Trinidad	Clerk	Purchasing	4F Rm 413

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Problem 2. LEFT JOIN

Display all employees and their department names.(employeeID, Firtname, Lastname and Department Name) Employees without a matching department must still appear in the result.

```
select e.EMP_ID , e.EMP_FIRSTNAME, e.EMP_LASTNAME , d.DEPT_NAME
from TB_EMPLOYEE e LEFT outer JOIN TB_DEPARTMENT d
on (e.DEPT_ID = d.DEPT_ID);
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	DEPT_NAME
222220	Rashley	Torres	Accounting
222224	Marris	Del Rosario	Accounting
222225	Alex	Causaren	Accounting
222227	Aaron	Sabater	Accounting
222221	Joanna	Mayuga	Marketing
222230	Jeffrey	Romero	Marketing
222231	Martheena	Herrera	Marketing
222232	Martin	Mesias	Marketing

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Problem 3. RIGHT JOIN

Display all departments and their employees. .(employeeID, Firtname, Lastname and Department No, Department Name)Departments without employees must appear even if the employee columns contain NULL.

```
1   select e.EMP_ID , e.EMP_FIRSTNAME, e.EMP_LASTNAME , d.DEPT_NAME  , d.DEPT_ID
2   from TB_EMPLOYEE e right outer JOIN TB_DEPARTMENT d
3   on (e.DEPT_ID = d.DEPT_ID);
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	DEPT_NAME	DEPT_ID
222221	Joanna	Mayuga	Marketing	888880
222232	Martin	Mesias	Marketing	888880
222231	Martheena	Herrera	Marketing	888880
222230	Jeffrey	Romero	Marketing	888880
222224	Marris	Del Rosario	Accounting	999990
222220	Rashley	Torres	Accounting	999990
222225	Alex	Causaren	Accounting	999990
222227	Aaron	Sabater	Accounting	999990

16 rows returned in 0.00 seconds Download

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Problem 4. FULL OUTER JOIN

Display all employees and all departments, including unmatched employees and departments. .(employeeID, Firtname, Lastname and Department No, Department Name) Identify which records have no matching department or employee.

```
1   select e.EMP_ID , e.EMP_FIRSTNAME, e.EMP_LASTNAME , d.DEPT_NAME  , d.DEPT_ID
2   from TB_EMPLOYEE e full outer JOIN TB_DEPARTMENT d
3   on (e.DEPT_ID = d.DEPT_ID);
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	DEPT_NAME	DEPT_ID
222220	Rashley	Torres	Accounting	999990
222221	Joanna	Mayuga	Marketing	888880
222222	Martha	Rosas	Purchasing	777770
222223	Marissa	Abad	Sales	666660
222224	Marris	Del Rosario	Accounting	999990
222225	Alex	Causaren	Accounting	999990
222226	Alley	Reyes	Payroll	555550
222227	Aaron	Sabater	Accounting	999990

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Activity B — Filtering and Sorting Joined Data

Problem 5. IT Department Employees

Display the full name, job title, salary, and department name of employees assigned to the Information Technology department. Sort the results by salary from highest to lowest.

```
1  SELECT e.EMP_ID, e.EMP_FIRSTNAME || ' ' || e.EMP_LASTNAME AS fullName,  e.POSITION, d.DEPT_NAME
2  FROM TB_EMPLOYEE e LEFT OUTER JOIN TB_DEPARTMENT d
3    ON (e.DEPT_ID = d.DEPT_ID)
4    order by e.SALARY DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	FULLNAME	POSITION	DEPT_NAME
222226	Alley Reyes	Manager	Payroll
222221	Joanna Mayuga	Manager	Marketing
222225	Alex Causaren	Clerk	Accounting
222220	Rashley Torres	Manager	Accounting
222224	Marris Del Rosario	Analyst	Accounting
222223	Marissa Abad	Manager	Sales
222222	Martha Rosas	Manager	Purchasing
222233	Arnie Agtonton	Analyst	Purchasing

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Problem 6. High-Salary Employees

Display the employee name, position, salary, and department name of employees earning more than ₱40,000. Include only employees with a valid department.

```
1   SELECT e.EMP_FIRSTNAME || ' ' || e.EMP_LASTNAME AS fullName,  e.POSITION, d.DEPT_NAME, salary
2   FROM TB_EMPLOYEE e inner JOIN TB_DEPARTMENT d
3   ON (e.DEPT_ID = d.DEPT_ID)
4   WHERE e.SALARY > 40000;
5
```

Results | Explain | Describe | Saved SQL | History

FULLNAME	POSITION	DEPT_NAME	SALARY
Valentino Saez	Supervisor	Payroll	45000
Alley Reyes	Manager	Payroll	55000
Marissa Abad	Manager	Sales	53000
Arnie Agtonton	Analyst	Purchasing	50000
Martha Rosas	Manager	Purchasing	53000
Martin Mesias	Analyst	Marketing	48000
Joanna Mayuga	Manager	Marketing	54000
Alex Causaren	Clerk	Accounting	54000
Rashley Torres	Manager	Accounting	53000
Marris Del Rosario	Analyst	Accounting	53000

10 rows returned in 0.01 seconds Download

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Problem 7. Employees Without a Valid Department

Display the employee ID, full name, and department ID of employees who either have no department assignment or have a department ID that does not exist in the DEPARTMENTS table.

```
1   SELECT e.EMP_ID, e.DEPT_ID, e.EMP_FIRSTNAME || ' ' || e.EMP_LASTNAME AS fullName
2   FROM TB_EMPLOYEE e LEFT OUTER JOIN TB_DEPARTMENT d
3   ON (e.DEPT_ID = d.DEPT_ID)
4   WHERE d.DEPT_ID IS NULL;
5
```

Results | Explain | Describe | Saved SQL | History

no data found

Problem 8. Departments Without Employees

Display the department ID, department name, and location of departments that currently have no employees.

```
1   SELECT d.DEPT_ID, d.DEPT_NAME, d.LOCATION
2   FROM TB_DEPARTMENT d LEFT OUTER JOIN TB_EMPLOYEE e
3   ON (d.DEPT_ID = e.DEPT_ID)
4   WHERE e.EMP_ID IS NULL;
```

Results | Explain | Describe | Saved SQL | History

no data found

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Problem 9. Average Salary per Department

Display the department name, number of employees, and average salary for each department.

Requirements:

 Include departments with no employees.

 Sort by average salary from highest to lowest, placing departments without salaries last.

```
1  SELECT d.DEPT_NAME, COUNT(e.EMP_ID) AS num_employees, ROUND(AVG(e.SALARY), 2) AS avg_salary
2  FROM TB_DEPARTMENT d LEFT OUTER JOIN TB_EMPLOYEE e
3  ON (d.DEPT_ID = e.DEPT_ID)
4  GROUP BY d.DEPT_ID, d.DEPT_NAME
5  ORDER BY avg_salary DESC NULLS LAST;
```

Results | Explain | Describe | Saved SQL | History

DEPT_NAME	NUM_EMPLOYEES	AVG_SALARY
Payroll	2	50000
Accounting	4	46375
Sales	2	41500
Marketing	4	37625
Purchasing	4	37250

5 rows returned in 0.01 seconds Download

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Problem 10. Employees Earning Above Their Department's Average Salary

Display the employee's full name, department name, salary, and department average salary for employees whose salary is greater than the average salary of employees in their own department.

```sql
SELECT e.EMP_FIRSTNAME || ' ' || e.EMP_LASTNAME AS fullName, d.DEPT_NAME, e.SALARY, ROUND(a.avg_sal, 2) AS dept_avg_salary
FROM TB_EMPLOYEE e INNER JOIN TB_DEPARTMENT d
ON (e.DEPT_ID = d.DEPT_ID)
INNER JOIN (
    SELECT DEPT_ID, AVG(SALARY) AS avg_sal
    FROM TB_EMPLOYEE
    GROUP BY DEPT_ID
) a
ON (e.DEPT_ID = a.DEPT_ID)
WHERE e.SALARY > a.avg_sal
ORDER BY d.DEPT_NAME, e.SALARY DESC;
```

Results | Explain | Describe | Saved SQL | History

FULLNAME	DEPT_NAME	SALARY	DEPT_AVG_SALARY
Alex Causaren	Accounting	54000	46375
Rashley Torres	Accounting	53000	46375
Marris Del Rosario	Accounting	53000	46375
Joanna Mayuga	Marketing	54000	37625
Martin Mesias	Marketing	48000	37625
Alley Reyes	Payroll	55000	50000
Martha Rosas	Purchasing	53000	37250
Arnie Agtonton	Purchasing	50000	37250
Marissa Abad	Sales	53000	41500

9 rows returned in 0.02 seconds Download

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201BA

ITC C201

Advanced Database / Oracle SQL

Sorting Records and Joining Tables

Problem 1 – Multi-level sorting

```
1  SELECT emp_id, EMP_FIRSTNAME, EMP_LASTNAME, position, salary, dept_id
2  FROM tb_employee
3  ORDER BY position ASC, salary DESC, EMP_LASTNAME ASC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	EMP_FIRSTNAME	EMP_LASTNAME	POSITION	SALARY	DEPT_ID
222224	Marris	Del Rosario	Analyst	53000	999990
222233	Arnie	Agtonton	Analyst	50000	777770
222232	Martin	Mesias	Analyst	48000	888880
222225	Alex	Causaren	Clerk	54000	999990
222228	Joseph	Trinidad	Clerk	19000	777770
222230	Jeffrey	Romero	Clerk	18500	888880
222226	Alley	Reyes	Manager	55000	555550
222221	Joanna	Mayuga	Manager	54000	888880
222223	Marissa	Abad	Manager	53000	666660
222222	Martha	Rosas	Manager	53000	777770
222220	Rashley	Torres	Manager	53000	999990
222231	Martheena	Herrera	Salesman	30000	888880
222234	Mark	Redublo	Salesman	30000	666660
222229	Christian	Delo Santos	Salesman	27000	777770
222227	Aaron	Sabater	Salesman	25500	999990
222235	Valentino	Saez	Supervisor	45000	555550

16 rows returned in 0.00 seconds Download

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Problem 2 – Salary ranking

```
1  SELECT emp_id, EMP_FIRSTNAME || ' ' || EMP_LASTNAME AS "Full Name", salary, emp_hiredate
2  FROM tb_employee
3  WHERE salary >= 40000
4  ORDER BY salary DESC, emp_hiredate ASC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	Full Name	SALARY	EMP_HIREDATE
222226	Alley Reyes	55000	4/3/1994
222225	Alex Causaren	54000	8/2/1993
222221	Joanna Mayuga	54000	3/15/1996
222222	Martha Rosas	53000	5/30/1997
222220	Rashley Torres	53000	2/25/1999
222223	Marissa Abad	53000	2/25/1999
222224	Marris Del Rosario	53000	9/11/1999
222233	Arnie Agtonton	50000	5/21/1994
222232	Martin Mesias	48000	9/11/2000
222235	Valentino Saez	45000	12/19/2000

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Problem 3 – Department and hire-date ordering

```
1  SELECT emp_id, EMP_FIRSTNAME || ' ' || EMP_LASTNAME AS "Full Name", dept_id, emp_hiredate
2  FROM tb_employee
3  WHERE emp_hiredate >= DATE '2000-01-01'
4    AND emp_hiredate <  DATE '2006-01-01'
5  ORDER BY dept_id ASC, emp_hiredate DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	Full Name	DEPT_ID	EMP_HIREDATE
222235	Valentino Saez	555550	12/19/2000
222234	Mark Redublo	666660	4/16/2001
222229	Christian Delo Santos	777770	10/23/2005
222228	Joseph Trinidad	777770	7/26/2001
222230	Jeffrey Romero	888880	10/26/2000
222232	Martin Mesias	888880	9/11/2000
222227	Aaron Sabater	999990	3/12/2000

7 rows returned in 0.01 seconds Download

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Problem 4 – Conditional sorting

```
SELECT emp_id, EMP_FIRSTNAME || ' ' || EMP_LASTNAME AS "Full Name", position, salary, emp_hiredate
FROM tb_employee
ORDER BY CASE position
WHEN 'Manager'    THEN 1
WHEN 'Supervisor' THEN 2
WHEN 'Analyst'    THEN 3
WHEN 'Salesman'   THEN 4
ELSE 5
END,
salary DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	Full Name	POSITION	SALARY	EMP_HIREDATE
222226	Alley Reyes	Manager	55000	4/3/1994
222221	Joanna Mayuga	Manager	54000	3/15/1996
222220	Rashley Torres	Manager	53000	2/25/1999
222222	Martha Rosas	Manager	53000	5/30/1997
222223	Marissa Abad	Manager	53000	2/25/1999
222235	Valentino Saez	Supervisor	45000	12/19/2000
222224	Marris Del Rosario	Analyst	53000	9/11/1999
222233	Arnie Agtonton	Analyst	50000	5/21/1994
222232	Martin Mesias	Analyst	48000	9/11/2000
222234	Mark Redublo	Salesman	30000	4/16/2001
222231	Martheena Herrera	Salesman	30000	11/22/1999
222229	Christian Delo Santos	Salesman	27000	10/23/2005
222227	Aaron Sabater	Salesman	25500	3/12/2000
222225	Alex Causaren	Clerk	54000	8/2/1993
222228	Joseph Trinidad	Clerk	19000	7/26/2001
222230	Jeffrey Romero	Clerk	18500	10/26/2000

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Problem 5 – Basic inner join

```
1  SELECT e.emp_id, e.EMP_FIRSTNAME || ' ' || e.EMP_LASTNAME AS "Full Name",
2         e.position, e.salary, d.dept_name, d.location
3  FROM tb_employee e
4  INNER JOIN tb_department d ON e.dept_id = d.dept_id
5  ORDER BY d.dept_name ASC, e.salary DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	Full Name	POSITION	SALARY	DEPT_NAME	LOCATION
222225	Alex Causaren	Clerk	54000	Accounting	3F Rm 301
222220	Rashley Torres	Manager	53000	Accounting	3F Rm 301
222224	Marris Del Rosario	Analyst	53000	Accounting	3F Rm 301
222227	Aaron Sabater	Salesman	25500	Accounting	3F Rm 301
222221	Joanna Mayuga	Manager	54000	Marketing	4F Rm 411
222232	Martin Mesias	Analyst	48000	Marketing	4F Rm 411
222231	Martheena Herrera	Salesman	30000	Marketing	4F Rm 411
222230	Jeffrey Romero	Clerk	18500	Marketing	4F Rm 411
222226	Alley Reyes	Manager	55000	Payroll	2F Rm 201
222235	Valentino Saez	Supervisor	45000	Payroll	2F Rm 201
222222	Martha Rosas	Manager	53000	Purchasing	4F Rm 413
222233	Arnie Agtonton	Analyst	50000	Purchasing	4F Rm 413
222229	Christian Delo Santos	Salesman	27000	Purchasing	4F Rm 413
222228	Joseph Trinidad	Clerk	19000	Purchasing	4F Rm 413
222223	Marissa Abad	Manager	53000	Sales	3F Rm 312
222234	Mark Redublo	Salesman	30000	Sales	3F Rm 312

16 rows returned in 0.01 seconds Download

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Problem 6 – Department salary analysis

```
1  SELECT d.dept_id, d.dept_name,
2  COUNT(e.emp_id) AS "No. of Employees",
3  AVG(e.salary) AS "Average Salary"
4  FROM tb_department d
5  INNER JOIN tb_employee e ON d.dept_id = e.dept_id
6  GROUP BY d.dept_id, d.dept_name
7  HAVING AVG(e.salary) > 35000
8  ORDER BY AVG(e.salary) DESC;
```

Results | Explain | Describe | Saved SQL | History

DEPT_ID	DEPT_NAME	No. of Employees	Average Salary
555550	Payroll	2	50000
999990	Accounting	4	46375
666660	Sales	2	41500
888880	Marketing	4	37625
777770	Purchasing	4	37250

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Problem 7 – Department-specific employee ranking

```
1  SELECT d.dept_name, e.emp_firstname || ' ' || e.emp_lastname AS "Full Name", e.position, e.salary
2  FROM tb_employee e
3  INNER JOIN tb_department d ON e.dept_id = d.dept_id
4  ORDER BY d.dept_name ASC, e.salary DESC, e.emp_lastname ASC;
```

Results | Explain | Describe | Saved SQL | History

DEPT_NAME	Full Name	POSITION	SALARY
Accounting	Alex Causaren	Clerk	54000
Accounting	Marris Del Rosario	Analyst	53000
Accounting	Rashley Torres	Manager	53000
Accounting	Aaron Sabater	Salesman	25500
Marketing	Joanna Mayuga	Manager	54000
Marketing	Martin Mesias	Analyst	48000
Marketing	Martheena Herrera	Salesman	30000
Marketing	Jeffrey Romero	Clerk	18500
Payroll	Alley Reyes	Manager	55000
Payroll	Valentino Saez	Supervisor	45000
Purchasing	Martha Rosas	Manager	53000
Purchasing	Arnie Agtonton	Analyst	50000
Purchasing	Christian Delo Santos	Salesman	27000
Purchasing	Joseph Trinidad	Clerk	19000
Sales	Marissa Abad	Manager	53000
Sales	Mark Redublo	Salesman	30000

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Problem 8 – Left join and unmatched departments

```
1  SELECT d.dept_id, d.dept_name, e.emp_id, e.emp_firstname || ' ' || e.emp_lastname AS "Full Name", e.salary
2  FROM tb_department d
3  LEFT JOIN tb_employee e ON d.dept_id = e.dept_id
4  ORDER BY d.dept_id, e.emp_lastname;
```

Results | Explain | Describe | Saved SQL | History

DEPT_ID	DEPT_NAME	EMP_ID	Full Name	SALARY
555550	Payroll	222226	Alley Reyes	55000
555550	Payroll	222235	Valentino Saez	45000
666660	Sales	222223	Marissa Abad	53000
666660	Sales	222234	Mark Redublo	30000
777770	Purchasing	222233	Arnie Agtonton	50000
777770	Purchasing	222229	Christian Delo Santos	27000
777770	Purchasing	222222	Martha Rosas	53000
777770	Purchasing	222228	Joseph Trinidad	19000
888880	Marketing	222231	Martheena Herrera	30000
888880	Marketing	222221	Joanna Mayuga	54000
888880	Marketing	222232	Martin Mesias	48000
888880	Marketing	222230	Jeffrey Romero	18500
999990	Accounting	222225	Alex Causaren	54000
999990	Accounting	222224	Marris Del Rosario	53000
999990	Accounting	222227	Aaron Sabater	25500
999990	Accounting	222220	Rashley Torres	53000

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```
1  SELECT d.dept_id, d.dept_name
2  FROM tb_department d
3  LEFT JOIN tb_employee e ON d.dept_id = e.dept_id
4  WHERE e.emp_id IS NULL;
```

Results | Explain | Describe | Saved SQL | History

no data found

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Problem 9 – Employees earning above their department average

```
1   SELECT e.emp_id, e.emp_firstname || ' ' || e.emp_lastname AS "Full Name", e.salary, d.dept_name,
2   da.avg_salary AS "Dept Avg Salary"
3   FROM tb_employee e
4   INNER JOIN tb_department d ON e.dept_id = d.dept_id
5   INNER JOIN (SELECT dept_id, AVG(salary) AS avg_salary
6   FROM tb_employee
7   GROUP BY dept_id) da
8   ON e.dept_id = da.dept_id
9   WHERE e.salary > da.avg_salary
10  ORDER BY d.dept_name, e.salary DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	Full Name	SALARY	DEPT_NAME	Dept Avg Salary
222225	Alex Causaren	54000	Accounting	46375
222220	Rashley Torres	53000	Accounting	46375
222224	Marris Del Rosario	53000	Accounting	46375
222221	Joanna Mayuga	54000	Marketing	37625
222232	Martin Mesias	48000	Marketing	37625
222226	Alley Reyes	55000	Payroll	50000
222222	Martha Rosas	53000	Purchasing	37250
222233	Arnie Agtonton	50000	Purchasing	37250
222223	Marissa Abad	53000	Sales	41500

9 rows returned in 0.03 seconds Download

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Problem 10 – Highest-paid employee per department

```
SELECT d.dept_name, e.emp_firstname || ' ' || e.emp_lastname AS "Full Name", e.position, e.salary
FROM tb_employee e
INNER JOIN tb_department d ON e.dept_id = d.dept_id
WHERE e.salary = (SELECT MAX(x.salary)
FROM tb_employee x
WHERE x.dept_id = e.dept_id)
ORDER BY d.dept_name;
```

Results | Explain | Describe | Saved SQL | History

DEPT_NAME	Full Name	POSITION	SALARY
Accounting	Alex Causaren	Clerk	54000
Marketing	Joanna Mayuga	Manager	54000
Payroll	Alley Reyes	Manager	55000
Purchasing	Martha Rosas	Manager	53000
Sales	Marissa Abad	Manager	53000

5 rows returned in 0.02 seconds Download

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Problem 11 – Department salary distribution

```
SELECT d.dept_name,
COUNT(e.emp_id) AS "Employee Count",
SUM(e.salary) AS "Total Salary",
MIN(e.salary) AS "Min Salary",
MAX(e.salary) AS "Max Salary",
AVG(e.salary) AS "Avg Salary"
FROM tb_department d
INNER JOIN tb_employee e ON d.dept_id = e.dept_id
GROUP BY d.dept_name
HAVING COUNT(e.emp_id) >= 2 AND SUM(e.salary) > 80000
ORDER BY SUM(e.salary) DESC;
```

Results | Explain | Describe | Saved SQL | History

DEPT_NAME	Employee Count	Total Salary	Min Salary	Max Salary	Avg Salary
Accounting	4	185500	25500	54000	46375
Marketing	4	150500	18500	54000	37625
Purchasing	4	149000	19000	53000	37250
Payroll	2	100000	45000	55000	50000
Sales	2	83000	30000	53000	41500

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Problem 12 – Employees with salaries above the company average

```
SELECT e.emp_id, e.emp_firstname || ' ' || e.emp_lastname AS "Full Name",
e.position, e.salary, d.dept_name,
c.company_avg AS "Company Avg"
FROM tb_employee e
LEFT JOIN tb_department d ON e.dept_id = d.dept_id
CROSS JOIN (SELECT AVG(salary) AS company_avg FROM tb_employee) c
WHERE e.salary > c.company_avg
ORDER BY e.salary DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	Full Name	POSITION	SALARY	DEPT_NAME	Company Avg
222226	Alley Reyes	Manager	55000	Payroll	41750
222225	Alex Causaren	Clerk	54000	Accounting	41750
222221	Joanna Mayuga	Manager	54000	Marketing	41750
222220	Rashley Torres	Manager	53000	Accounting	41750
222224	Marris Del Rosario	Analyst	53000	Accounting	41750
222222	Martha Rosas	Manager	53000	Purchasing	41750
222223	Marissa Abad	Manager	53000	Sales	41750
222233	Arnie Agtonton	Analyst	50000	Purchasing	41750
222232	Martin Mesias	Analyst	48000	Marketing	41750
222235	Valentino Saez	Supervisor	45000	Payroll	41750

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Problem 13 – Department comparison

```
1   SELECT d.dept_name,
2   COUNT(e.emp_id) AS "Employee Count",
3   AVG(e.salary) AS "Dept Avg",
4   c.company_avg AS "Company Avg"
5   FROM tb_department d
6   INNER JOIN tb_employee e ON d.dept_id = e.dept_id
7   CROSS JOIN (SELECT AVG(salary) AS company_avg FROM tb_employee) c
8   GROUP BY d.dept_name, c.company_avg
9   HAVING AVG(e.salary) > c.company_avg
10  ORDER BY AVG(e.salary) - c.company_avg DESC;
```

Results | Explain | Describe | Saved SQL | History

DEPT_NAME	Employee Count	Dept Avg	Company Avg
Payroll	2	50000	41750
Accounting	4	46375	41750

2 rows returned in 0.01 seconds Download

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Problem 14 – Position-based salary analysis

```
1   SELECT d.dept_name, e.position,
2   COUNT(e.emp_id) AS "Employee Count",
3   AVG(e.salary) AS "Avg Salary"
4   FROM tb_employee e
5   INNER JOIN tb_department d ON e.dept_id = d.dept_id
6   GROUP BY d.dept_name, e.position
7   HAVING COUNT(e.emp_id) >= 2 AND AVG(e.salary) >= 30000
8   ORDER BY d.dept_name, AVG(e.salary) DESC;
```

Results | Explain | Describe | Saved SQL | History

no data found

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Problem 15 – Advanced conditional join and sorting

```
1  SELECT e.emp_id, e.emp_firstname || ' ' || e.emp_lastname AS "Full Name",
2  e.position, e.emp_hiredate, e.salary, d.dept_name, d.location
3  FROM tb_employee e
4  LEFT JOIN tb_department d ON e.dept_id = d.dept_id
5  WHERE (e.salary > 45000)
6  OR (e.position = 'Manager' AND e.salary >= 40000)
7  OR (e.emp_hiredate < DATE '2000-01-01' AND e.salary > 30000)
8  ORDER BY d.dept_name, e.salary DESC;
```

Results | Explain | Describe | Saved SQL | History

EMP_ID	Full Name	POSITION	EMP_HIREDATE	SALARY	DEPT_NAME	LOCATION
222225	Alex Causaren	Clerk	8/2/1993	54000	Accounting	3F Rm 301
222220	Rashley Torres	Manager	2/25/1999	53000	Accounting	3F Rm 301
222224	Marris Del Rosario	Analyst	9/11/1999	53000	Accounting	3F Rm 301
222221	Joanna Mayuga	Manager	3/15/1996	54000	Marketing	4F Rm 411
222232	Martin Mesias	Analyst	9/11/2000	48000	Marketing	4F Rm 411
222226	Alley Reyes	Manager	4/3/1994	55000	Payroll	2F Rm 201
222222	Martha Rosas	Manager	5/30/1997	53000	Purchasing	4F Rm 413
222233	Arnie Agtonton	Analyst	5/21/1994	50000	Purchasing	4F Rm 413
222223	Marissa Abad	Manager	2/25/1999	53000	Sales	3F Rm 312

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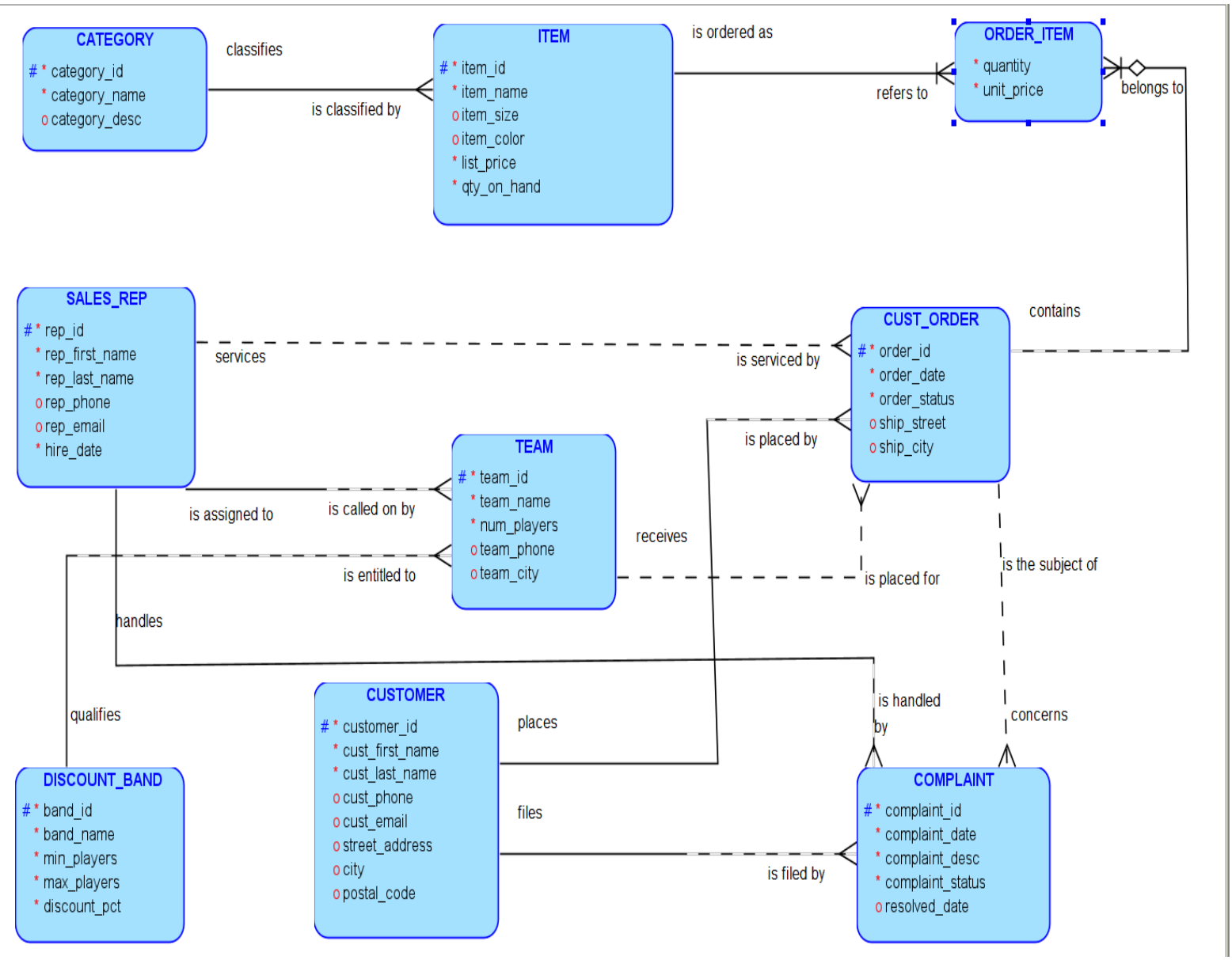
SECTION 7

Projects and Performance Tasks

Contents of this section

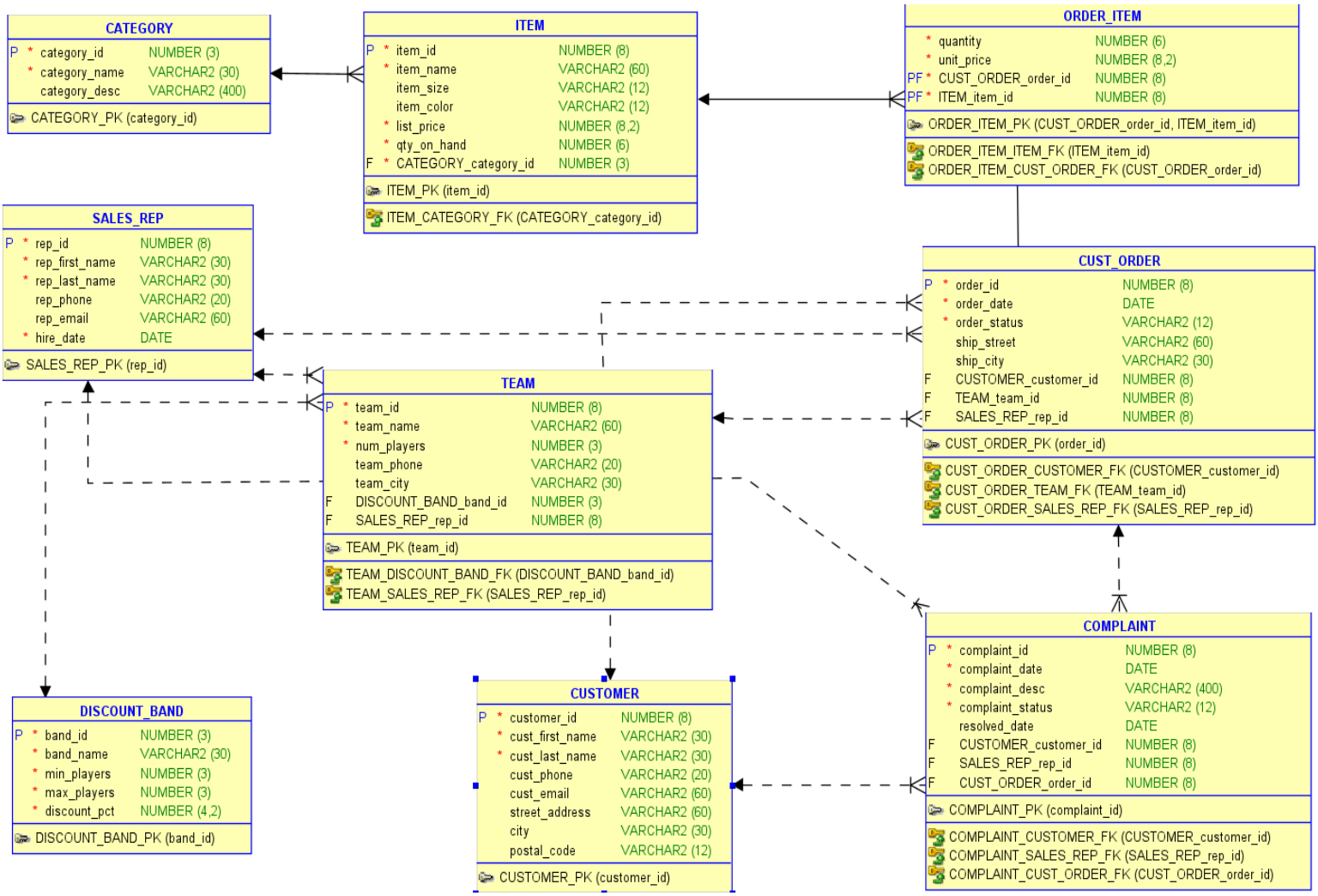
- Prelim Project – ERD and Normalization

ITC C201 PRELIM PROJECT - ERD and Normalization



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ITC C201 PRELIM PROJECT - ERD and Normalization



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8. Learning Reflection

What important knowledge and skills did I learn?

In this course I learned how a database is designed and how it is used. I learned to draw ERDs in SQL Data Modeler, including entities, relationships, and supertype and subtype entities, and to apply normalization. In Oracle APEX, I learned to create tables with data types, primary keys, foreign keys, NOT NULL, and UNIQUE constraints, and to change them with ALTER TABLE and DROP TABLE. I also learned to add, change, and delete records with INSERT, UPDATE, and DELETE, and to retrieve data with SELECT using aliases, arithmetic and concatenation, WHERE, LIKE, ORDER BY, DISTINCT, and aggregate functions such as MIN, MAX, SUM, AVG, and COUNT. Finally, I learned GROUP BY, HAVING, subqueries, and the different join types: INNER, LEFT, RIGHT, FULL OUTER, NATURAL, and USING.

What activities or topics were most challenging?

The most challenging topics were the outer joins and the queries that compare a value with an average, such as employees who earn more than their department average. It was hard to decide which table should be on the left or right side of a join, and when to use a subquery. Normalization and the supertype and subtype design in the ERD were also difficult at first.

How did I overcome the difficulties?

I practiced by running each query in Oracle APEX and checking the result against the data in the table. After every change I used DESCRIBE or SELECT * to confirm that the table was correct. When a statement gave an error, I read the message, fixed the statement, and ran it again; my SQL history shows that I repeated this many times. I also learned that a result with no data can still be correct. For example, the queries for employees without a valid department and for departments without employees returned no data because every record matched.

How will I apply the knowledge and skills I acquired?

As a Business Analytics student, I will use SQL to collect data from several tables, combine it with joins, and summarize it with aggregate functions for reports and analysis. I will also apply the CRUD and database design concepts in my other subjects and projects, such as my Flutter and Firebase app, and in my personal projects. These skills are a strong base for the advanced analytics and database courses that come next.

9. Course Learning Summary

ITC C201 – Database Fundamentals gave me a complete path from planning a database to querying it. In the Prelim period, I designed conceptual and logical models and an ERD with normalization. In the Midterm period, I created my Oracle APEX account, built tables, and inserted, updated, and deleted records. In the Final period, I worked with aliases, sorting, aggregate functions, and all major join types, and I solved multi-step problems with grouping and subqueries.

Overall, I completed every activity, laboratory exercise, and project that is included in this portfolio, and each one shows the SQL code and a screenshot of the actual output. I improved my accuracy, my patience in checking results, and my confidence in writing SQL. This portfolio shows my progress from the first ERD sketch to advanced join queries, and I will continue to improve my skills in the next courses.